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PROGRAMMING FOR DATA ANALYTICS 2	PDAN8412/w

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Part 2: Logistic Regression

Predicting best-selling books

A report to the publisher on data, method, model and findings

Dataset: Goodbooks-10k, 10,000 books from Goodreads (Zając, 2017)

Method: binary logistic regression written from scratch in Python (NumPy), with scikit-learn used only to split data, compute metrics and validate

Code: Part2_Logistic_Regression.ipynb (executed notebook) and part2_logistic_regression.py

Referencing style: Harvard (Anglia)

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Executive summary

The ROC-AUC of the logistic regression fit from the beginning is 0.81. In the end, it identifies 57.5% of the books that are actually best sellers, while 53.5% of the books it alerts are actual best sellers.

What we did. For this purpose, we have selected 10,000 popular books from Goodreads (Zając, 2017). The top 20% by number of reader ratings (80th percentile of 55,674 ratings) is considered a best seller. We cleaned and explored the data, checked the assumptions of logistic regression, selected 11 predictors based on p-values, trained the model with our own NumPy implementation, tested it on 2000 books that it had never seen before, and finally retrained our model to address its primary fault.

Metric (2,000 unseen books)	Model 1 (first attempt)	Model 2 (final)
Accuracy	0.844	0.815
Precision (flagged books that are best sellers)	0.754	0.535
Recall (best sellers found)	0.323	0.575
F1 score	0.452	0.554
ROC-AUC (ranking quality)	0.813	0.813

Table 1: Headline results on the test set

What this means for the publisher

- Model's results were reasonably good (84%) with only 2 out of 3 best sellers incorrect. For each book, the answer “not a best seller” gets 80% correct, which is misleading by itself (Saito and Rehmsmeier, 2015).
- Model 2 is not only as powerful as Model 1 but also has a better decision point. It finds 57.5% of best sellers instead of 32.3% (F1 rises from 0.452 to 0.554). The cost is more false alarms.
- The top 20% of books in terms of scores yield 55% of all best sellers, which is a boost of 2.75 over random selection.
- The most powerful are the number of editions, the average rating, the fact that it is the first book of a series and the number of books the author has.
- Caution: If there were no number of editions, which in part shows success already achieved, then AUC drops to 0.67. Do not use the model as a forecast..

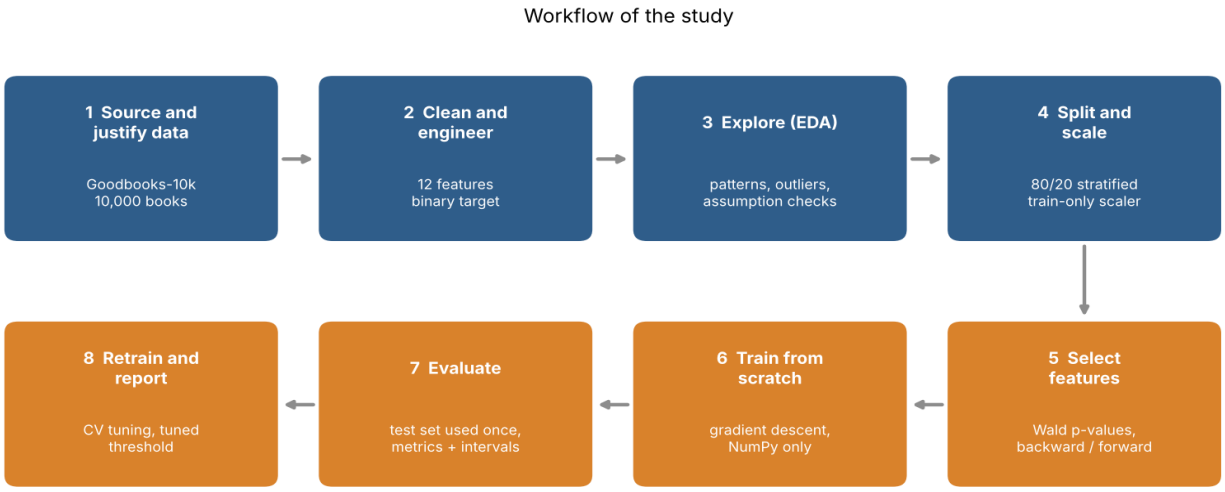


Figure 1: Workflow of the study

1 Introduction and business context

A publishing company wishes to determine, at an early stage, whether it is likely that any of its books will become best sellers. This information would assist it in determining what books they would print, promote or translate. The publisher requested a model to predict a yes/no response (best seller or not) and for an explanation of how the model was created and how much it can be trusted.

Logistic regression is the obvious first model to try when the outcome is a yes/no result. It is easy to handle, coefficients can be interpreted as odds ratios and it has a long history of application in practice (Hosmer, Lemeshow and Sturdivant, 2013; James et al., 2021). The algorithm description requires that the algorithm is implemented from scratch, indicating that each step is known, not in a library.

Research question. Is it possible to build a logistic regression model using publicly-available data on a book and predict best-sellers accurately enough for a book publisher to use them to select books to include in its shortlist? What features are important in the predictions of their logistic regression model?

Objectives

1. Source a dataset and justify its quality and suitability for logistic regression.
2. Design the analysis before conducting the analysis.
3. Purify the data, delve into the data and examine the assumption of the method.
4. Pick features based on p-values (backward and forward).
5. Train a logistic regression from scratch and explain all of the steps.
6. Test, analyse and understand results and then retrain to correct for any deficiencies.
7. Report the process and results in a format that is suitable for the client..

Brief / rubric item	Where it is answered
Task 1: evaluate and justify the dataset (rubric: justification of suitability, 10 marks)	Sections 3 and 5
Task 2: plan the analysis, parts a to e (rubric: planning analysis, 20 marks)	Section 4

Brief / rubric item	Where it is answered
Task 3a: exploratory data analysis (rubric: thorough EDA in Python, 20 marks)	Sections 5 and 6; notebook sections 2, 3, 3.9 and 3b
Task 3b: feature selection with p-values	Sections 7 and 8; notebook sections 6 and 6b
Task 3c: train the model (rubric: model trained in Python, 10 marks)	Section 9; notebook sections 5, 7 and 8
Task 4a: interpret and evaluate (rubric: evaluated and interpreted using Python libraries, 20 marks)	Sections 8 and 10; notebook sections 9, 9.9 and 9b
Task 4b: retrain with different parameters	Section 11; notebook section 8
Task 5: write and submit the report (rubric: report, 20 marks)	This document

Table 2: How this report answers the brief and the marking rubric

The report is presented in the order of the work: theory (section 2), data (3-6), modelling (7-9), evaluation and retraining (10, 11) and discussion (12-14). The following text is a description in plain words of snippets of Python code; the full code with comments is in the notebook.

2 Background and theory

2.1 Why logistic regression for a yes/no outcome

If the outcome is a 0/1 binary classification, a linear regression fitted to that outcome can predict values less than 0 or greater than 1, and the errors of this linear regression do not need to be normally distributed with constant variance. Cox (1958) introduced logistic regression for binary sequences, which solves both problems by modelling the probability of the outcome with an S-shaped curve that is bounded by the interval [0, 1] (Hosmer, Lemeshow and Sturdivant, 2013; James et al., 2021).

2.2 The model: odds, log-odds and the sigmoid

The odds that a book is a best seller are equal to p / (1 − p), where p is the probability that a book is a best seller. Logistic regression makes the assumption that the log-odds are a linear function of features. The reverse of that is the sigmoid function:

$$\ln\left(\frac{p}{1-p}\right) = b_0 + b_1 x_1 + b_2 x_2 + \ldots + b_k x_k = z$$

$$p = \frac{1}{1 + e^{-z}}$$

Each coefficient is an actual value: a change of one unit in a feature results in a change of the odds by e to the power of the coefficient (the odds ratio). If the odds ratio is 1.5, then there are 50% greater odds, and if it is 0.5, there are half the odds (Hosmer, Lemeshow and Sturdivant, 2013).

2.3 Estimation: maximum likelihood, gradient descent and Newton-Raphson

The coefficients are such as to maximize the likelihood that the observed outcomes will occur. They, in turn, minimize the binary cross-entropy (log-loss) which is heavily penalized for confident mis-predictions:

$$L = -\frac{1}{n} \sum \left[y \cdot \ln(p) + (1 - y) \cdot \ln(1 - p) \right]$$

This This loss is convex, meaning that it is unimodal with no false valleys in which to trap (Bishop, 2006; Hastie, Tibshirani and Friedman, 2009). There is no closed form solution and it is solved iteratively. Two methods are used in this study:

In the case of gradient descent, the coefficients are taken a small step downhill again and again. It has a simple form, which is the average of (p − y) times each feature. The step size is the learning rate. The models in this report are trained using this method.

This method relies on the curvature of the loss too and so is converged in a few steps, Newton-Raphson (iteratively re-weighted least squares). It also yields its inverse, which is the same as the standard errors required for p-values (Hastie, Tibshirani and Friedman, 2009).

Newton step: b_new = b + (X'WX)^-1 · X'(y - p), W = diag(p(1 - p))

Wald statistic: z_j = b_j / SE(b_j), p-value from the normal curve

2.4 Regularisation, class imbalance and the decision threshold

The L2 (ridge) penalty is a term that is proportional to the squared coefficients and is added to the loss; this causes the coefficients to be smaller and towards zero, preventing over-fitting. Errors on the rare class count more in class weights. A third and simpler approach is to leave the model as is, but shift the decision threshold off of 0.5. He and Garcia (2009) review the following remedies for imbalanced data, this study experiments with all three.

2.5 Evaluating a classifier when one class is rare

If 80% of the cases in one class, a model that always predicts class A is 80% accurate, this tells little about the accuracy. The study, therefore, provides several measures, each addressing a different question (Sokolova and Lapalme, 2009):

Precision = TP / (TP + FP) Recall = TP / (TP + FN) F1 = 2·P·R / (P + R)

MCC = (TP·TN - FP·FN) / sqrt((TP+FP)(TP+FN)(TN+FP)(TN+FN))

ROC-AUC is the chance that a randomly chosen best seller is scored above a randomly chosen non-best-seller; 0.5 is guessing (Fawcett, 2006).

- PR-AUC (average precision) is based on the rare class, and it has a baseline equal to the percentage of positives. When the number of positives is small, the precision-recall analysis is more informative than ROC (Davis and Goadrich, 2006; Saito and Rehmsmeier, 2015).
- The Matthews correlation coefficient (MCC) considers all four cells in the confusion matrix and maintains honesty on imbalanced data (Chicco and Jurman, 2020).
- To ask such questions as whether a predicted 30% really indicates approximately 30% of these books are best sellers is known as calibration (Niculescu-Mizil and Caruana, 2005). It is summarised by the Brier score, which is the mean squared gap between probability and outcome (Brier, 1950).
- Massey (1951) proposed the Kolmogorov-Smirnov (KS) statistic, which is the largest difference between the score distributions of the two classes.

2.6 Choosing features, sample size and checking the model

Backward elimination (left with all, remove the weakest) or forward selection (remove all, add the strongest) are both ways of selecting features based on p-values. With many features being tested simultaneously, a more stringent cutoff is used to prevent false positives (Bonferroni correction: Dunn, 1961). At least 10 events per variable, as per a common guide (Peduzzi et al., 1996). Collinearity is assessed with the variance inflation factor (VIF); O'Brien (2007) warns against setting a hard cutoff of any magnitude but a value > 5 or 10 is typically considered alarming. The study applies the likelihood-ratio test and pseudo-R-squared (McFadden 1974; Nagelkerke 1991), Hosmer-Lemeshow goodness-of-fit test (Hosmer and Lemeshow 1980) and Cook's distance for influential cases (Cook 1977).

2.7 Validation

A model must be evaluated when it encounters data for which it wasn't trained. The study has a test set stored in a separate file until the very end, and then tunes everything else on the training set using k-fold cross-validation (Kohavi, 1995), with a confidence interval around the final score using the bootstrap (Efron and Tibshirani, 1993). Throughout, exploratory data analysis is employed, in the spirit of Tukey (1977), which involves examining the data before modelling it.

2.8 Tools

The data set is Goodbooks-10k (Zając, 2017), comprising of 10,000 popular books from Goodreads (2026). Each row is one book with its authors, first publication year, language, title, number of editions, average rating and counts of ratings. There are 23 columns in the file, and it was downloaded and used unchanged as books.csv. Table 2 explains all of the columns and what they are used for.

3 The dataset: source, quality and suitability

3.1 Source and description

The data set is Goodbooks-10k (Zając, 2017), comprising of 10,000 popular books from Goodreads (2026). Each row is one book with its authors, first publication year, language, title, number of editions, average rating and counts of ratings. There are 23 columns in the file, and it was downloaded and used unchanged as books.csv. Table 2 explains all of the columns and what they are used for.

Column(s)	Type	Meaning	Use in this study
book_id, goodreads_book_id, best_book_id, work_id	integer	Identifiers used by the dataset and by Goodreads	Not used: identifiers carry no predictive meaning
books_count	integer	Number of editions of the work	Used: log_books_count
isbn, isbn13	text, decimal	ISBN-10 and ISBN-13 numbers	isbn becomes the has_isbn flag; isbn13 not used
authors	text	Author and contributor names, comma-separated	Used: n_authors, log_author_titles
original_publication_year	decimal	Year the work was first published	Used: log_book_age
original_title, title	text	Original title; title including series information	title gives is_series, is_series_book1, title_words, has_subtitle
language_code	text	Language of the edition (eng, en-US, fre, ...)	Used: lang_other, lang_unknown
average_rating	decimal	Mean star rating, 1 to 5	Used as it is
ratings_count, work_ratings_count	integer	Number of ratings	work_ratings_count defines the target; neither is a feature (leakage)
work_text_reviews_count	integer	Number of written reviews	Excluded: measures the same popularity as the target

Column(s)	Type	Meaning	Use in this study
ratings_1 to ratings_5	integer	Ratings at each star level	Excluded: they add up to the target; used only in one descriptive chart
image_url, small_image_url	text	Links to cover images	Not used

Table 3: Data dictionary for the raw file

3.2 Why the dataset is suitable

Requirement of logistic regression	How the dataset meets it
A binary outcome	Best seller = top 20% of books by number of ratings: 2,000 of 10,000 books. Sales figures are not public, so ratings act as a popularity proxy.
Enough events	2,000 best sellers. The guide of 10 events per variable (Peduzzi et al., 1996) needs only about 120; the study has 145.45 per variable in the training data.
Independent rows	One row per book; no duplicate book_id or work_id.
Mixed predictor types	Numeric (rating, editions, year), yes/no and categorical (language, series, subtitle), so encoding and selection are both meaningful.
No target leakage	ratings_count, work_ratings_count, ratings_1 to ratings_5 and work_text_reviews_count define or mirror the label, so they are not allowed as predictors.
Little collinearity	Every predictor has a VIF of 2.2 or less (section 6.8).
Log-odds linear in the features	Skewed variables are log-transformed (section 5), and decile plots confirm the pattern (section 6.8).

Table 4: Suitability of the dataset for logistic regression

3.3 Data quality audit

Check	Finding	Action
Missing values	Publication year 21 rows (0.21%), language code 1,084 (10.84%), ISBN 700 (7%); isbn13 and original_title 585 each (5.85%)	Median year; “unknown” language group; has_isbn flag; the other two columns are not used
Duplicates	None in book_id or work_id; 36 repeated titles are different works	Kept
Extreme values	Years back to -1750 (125 books before 1800 are genuine classics); up to 3,455 editions	Age capped at the 99th percentile, then log; log of editions
Consistency	Star counts add up exactly to the total ratings (maximum difference 0)	None needed
Class balance	80% not best seller, 20% best seller	Stratified split; F1, PR-AUC; tuned threshold

Check	Finding	Action
Hidden structure	Series information is written inside the title text	Parsed into series flags

Table 5: Data quality checks and the action taken

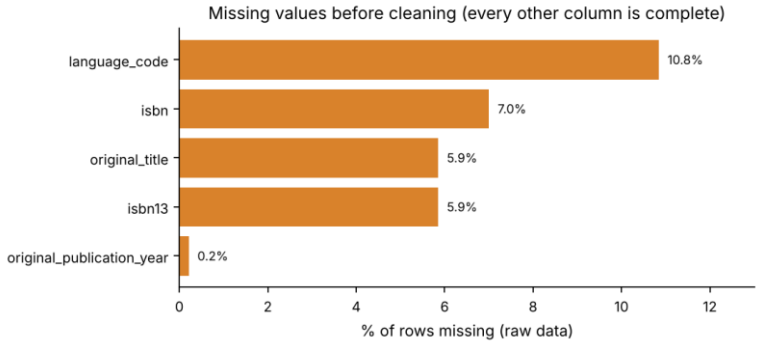


Figure 2: Share of rows missing in each column that has gaps

3.4 The target and the class balance

The target is defined from the data: a book is a best seller if its number of ratings (work_ratings_count) is above the 80th percentile, 55,674 ratings. This gives 2,000 best sellers out of 10,000. The use of a percentile instead of a fixed number will allow the classes to be stable, and the 80/20 imbalance will be a known characteristic of the problem.

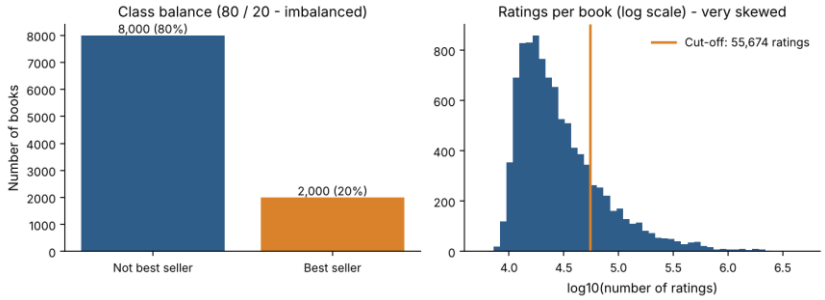


Figure 3: Class balance and the best-seller cut-off on the log scale

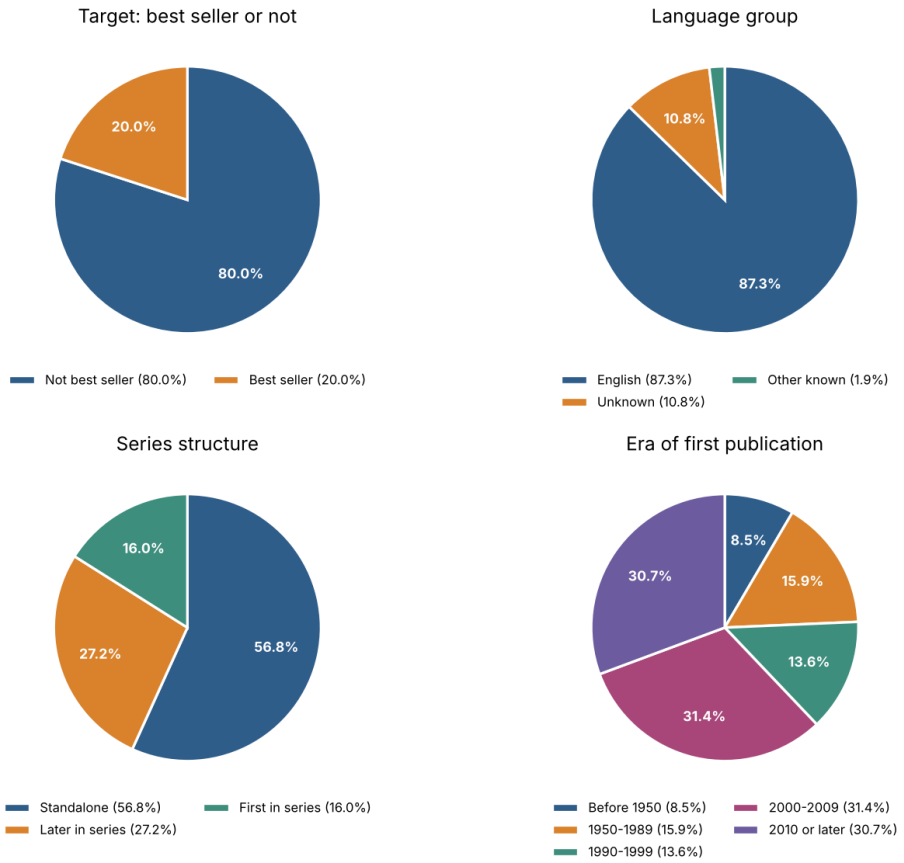


Figure 4: Composition of the dataset: target, language, series structure and era of first publication

3.5 Limits of the dataset

Popularity is just a proxy of sales and not everyone who reads on Goodreads is a book buyer! This is because each book is already popular; the model does not identify hits versus flops, but only very-popular versus moderately-popular books (selection bias). Success has led to a number of editions. The limits are tested or discussed in sections 10, 12 and 13.

4 Analysis plan

This plan was formulated before modelling was done. It has five steps, with each step having decisions and the rationale for those decisions. Success is defined by a ROC-AUC > 0.75, and a final model that performs better than the baseline on F1 score while maintaining ranking power.

Stage	What was planned	Why / extra consideration
a. Exploratory data analysis and cleaning	Audit missing values, duplicates, extremes and consistency, then fix them. Engineer 12 features. Study the target, each feature by class, correlations, trend over time, outliers (IQR rule), statistical tests, assumption checks and VIF.	Finds problems before they reach the model; decides the log transforms; shows the 80/20 imbalance early (Tukey, 1977).
b. Feature selection	Fit by Newton-Raphson to get Wald p-values. Backward elimination (start with all) and forward selection (start empty) at $p < 0.01$, with $p < 0.05$ as a check. Compare chosen sets by 5-fold cross-validation.	With 12 tests and 8,000 rows, $p < 0.05$ is too lenient (Bonferroni would be about 0.004; Dunn, 1961). Backward judges each feature given all the others.

Stage	What was planned	Why / extra consideration
c. Train model	Stratified 80/20 split (seed 42); scale with training data only. Model 1: learning rate 0.1, 1,000 iterations, no penalty, threshold 0.5. Model 2 grid: learning rate {0.01, 0.05, 0.1, 0.5} × L2 {0, 0.001, 0.01, 0.1} × class weights {none, balanced}, 5-fold CV; threshold from out-of-fold F1.	A plain baseline first gives something to beat. Tuning with cross-validation on the training set keeps the test set honest (Kohavi, 1995).
d. Interpret and evaluate	Accuracy, precision, recall, F1, ROC-AUC, PR-AUC, MCC, log-loss, Brier score, KS, calibration, gains and lift; bootstrap interval for AUC; odds ratios; permutation importance; subgroup analysis; comparison with scikit-learn; a run without the editions feature; error analysis.	Accuracy misleads at 80/20. The scikit-learn check proves the maths; the sensitivity run tests whether the model can be trusted before publication.
e. Report	Plain-language report of cleaning, EDA, selection, training, results and limits, with every figure reproducible from the code.	The client can follow every decision.

Table 6: The analysis plan

Analysis Plan

From data to insights | A step-by-step approach

1 Exploratory Data Analysis and Cleaning

What was planned:

- Audit missing values, duplicates, extremes and consistency, then fix them.
- Engineer 12 features.
- Study the target, each feature by class, correlations, trend over time, outliers (IQR rule), statistical tests, assumption checks and VIF.

Why / extra consideration:

Finds problems before they reach the model; decides the log transforms; shows the 80/20 imbalance early (Tukey, 1977).

2 Feature Selection

What was planned:

- Fit by Newton-Raphson to get Wald p-values.
- Backward elimination (start with all) and forward selection (start empty) at $p < 0.01$, with $p < 0.05$ as a check.
- Compare chosen sets by 5-fold cross-validation.

Why / extra consideration:

With 12 tests and 8,000 rows, $p < 0.05$ is too lenient (Bonferroni would be about 0.004; Dunn, 1961). Backward judges each feature given all the others.

3 Train Model

What was planned:

- Stratified 80/20 split (seed 42); scale with training data only.
- **Model 1:** learning rate 0.1, 1,000 iterations, no penalty, threshold 0.5.
- **Model 2 grid:** learning rate {0.01, 0.05, 0.1, 0.5} × L2 {0, 0.001, 0.01, 0.1} × class weights {none, balanced}, 5-fold CV; threshold from out-of-fold F1.

Why / extra consideration:

A plain baseline first gives something to beat. Tuning with cross-validation on the training set keeps the test set honest (Kohavi, 1995).

4 Interpret and Evaluate

What was planned:

- Accuracy, precision, recall, F1, ROC-AUC, PR-AUC, MCC, log-loss, Brier score, KS, calibration, gains and lift.
- Bootstrap interval for AUC.
- Odds ratios.
- Permutation importance.
- Subgroup analysis.
- Comparison with scikit-learn.
- A run without the editions feature.
- Error analysis.

Why / extra consideration:

Accuracy misleads at 80/20. The scikit-learn check proves the maths; the sensitivity run tests whether the model can be trusted before publication.

5 Report

What was planned:

- Plain-language report of cleaning, EDA, selection, training, results and limits, with every figure reproducible from the code.

Why / extra consideration:

The client can follow every decision.

5 Data cleaning and feature engineering

Every cleaning decision below is in response to a specific problem in the audit. The goal is to have a model-ready data table that has no missing values and where the features' scales and shapes are appropriate for logistic regression.

Problem	Decision	Reason
21 missing publication years (0.2%)	Fill with the median year	The median is not pulled by very old books. With so few gaps the effect is negligible (Little and Rubin, 2019).
1,084 missing language codes (10.8%)	Own category “unknown”	Only 7.4% of these books are best sellers versus 21.5% of the rest, so the gap itself carries information
Years back to -1750	Age = 2017 minus year, capped at the 99th percentile, then log	A few ancient books must not dominate
Editions up to 3,455	log(1 + editions)	Skewness falls from 8.41 to 0.47; logistic regression prefers symmetric inputs
Many language codes	English (reference), lang_other, lang_unknown	Collapses en-US, en-GB and so on, without a collinearity trap
Series inside the title text	is_series and is_series_book1	Series books are commercially different
Long titles, many credited names	Title words capped at 15; names capped at 5	Limits outliers
Prolific authors	log(1 + the author's titles in the data)	Measures an author's catalogue size
Missing ISBN	has_isbn flag	The ISBN itself is only an identifier

Table 7: Cleaning decisions and their reasons

Feature	Definition	Type
average_rating	Mean star rating	continuous
log_books_count	ln(1 + number of editions)	continuous
log_book_age	ln(1 + age in 2017), age capped at the 99th percentile	continuous
lang_other	1 if the language is known and not English	binary
lang_unknown	1 if the language code is missing	binary
has_isbn	1 if an ISBN is present	binary
is_series	1 if the title ends with “(Series, #n)”	binary
is_series_book1	1 if the book is number 1 in its series	binary
title_words	Words in the title without the series text (maximum 15)	count
has_subtitle	1 if the title contains a colon	binary

Feature	Definition	Type
n_authors	Credited names (maximum 5)	count
log_author_titles	ln(1 + titles by the first author in the dataset)	continuous

Table 8: The twelve engineered features

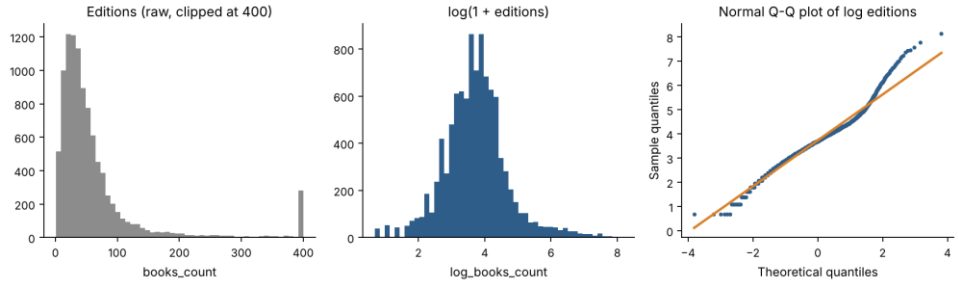


Figure 5: Editions before and after the log transform, with a normal Q-Q plot

The log transform does what it should. For editions, skewness falls from 8.41 to 0.47; for book age it falls from 4.04 to 0.5. The Q-Q plot hugs the line in the middle and bends only in the tails. The spike at 400 in the left histogram is a display artefact: values above 400 are drawn in that last bar.

Feature	Mean	Std. dev.	Minimum	Median	Maximum
average_rating	4.00	0.25	2.47	4.02	4.82
log_books_count	3.73	0.96	0.69	3.71	8.15
log_book_age	2.72	1.01	0.00	2.64	5.69
lang_other	0.02	0.14	0.00	0.00	1.00
lang_unknown	0.11	0.31	0.00	0.00	1.00
has_isbn	0.93	0.26	0.00	1.00	1.00
is_series	0.43	0.50	0.00	0.00	1.00
is_series_book1	0.16	0.37	0.00	0.00	1.00
title_words	3.89	2.87	1.00	3.00	15.00
has_subtitle	0.13	0.34	0.00	0.00	1.00
n_authors	1.29	0.66	1.00	1.00	5.00
log_author_titles	1.85	0.99	0.69	1.79	4.60

Table 9: Summary statistics of the model features after cleaning (10,000 books)

6 Exploratory data analysis

The exploration starts with the whole data set, progresses to individual features, to pairs of features, and finally to making assumptions about the method. A modelling decision is included in the summary at the end of the section for each finding.

6.1 Composition: who is in the data?

The dataset is 87.3% English-language books, but 95.5% of the best sellers. Books with unknown language are 10.8% of all books, but 4.0% of best sellers, and books known to be in another language are 1.9% of all books and 0.5% of best sellers. Of the data, 56.8% is in standalone books. The first books in book series constitute 16.0% of all books but 20.8% of best sellers. This data is current: 62.1% of the books were published in 2000 or later.

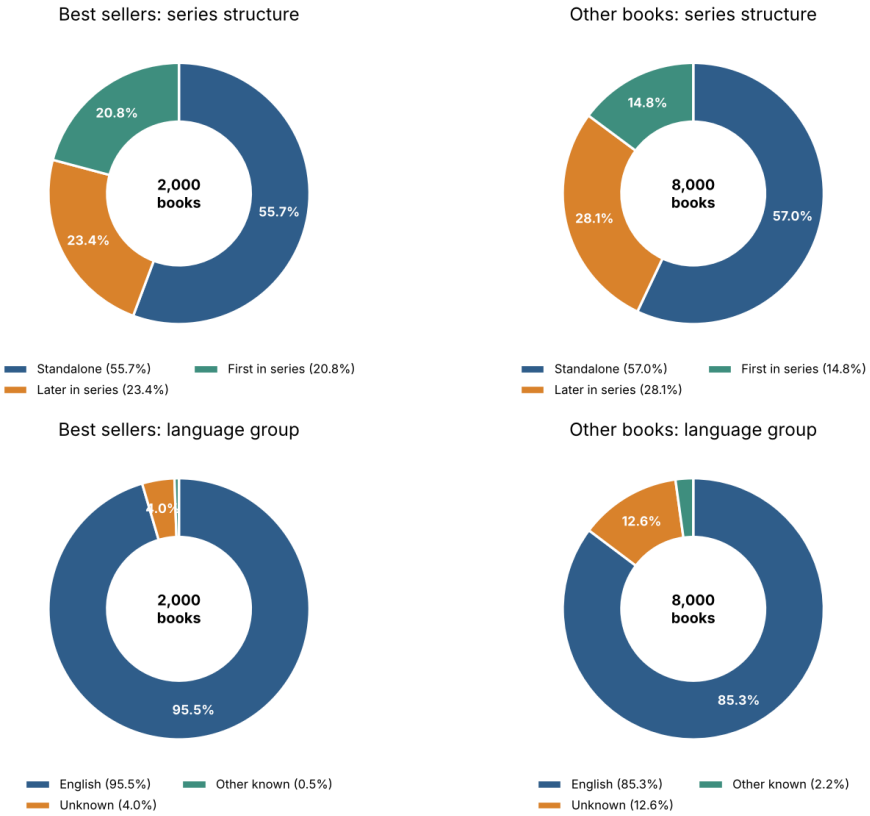


Figure 6: Series structure and language group among best sellers and other books

6.2 Distributions by class

The greatest distinction is in the number of editions. The median best seller has 76 editions, compared to 34 for other books (medians of 4.34 and 3.56 of log editions). The size of the author catalogues varies as well (median 2.08 against 1.61 on the log scale). Average rating barely changes (median 4.03 vs. 4.01) and the number of credited names and the length of the title are almost identical for the two groups.

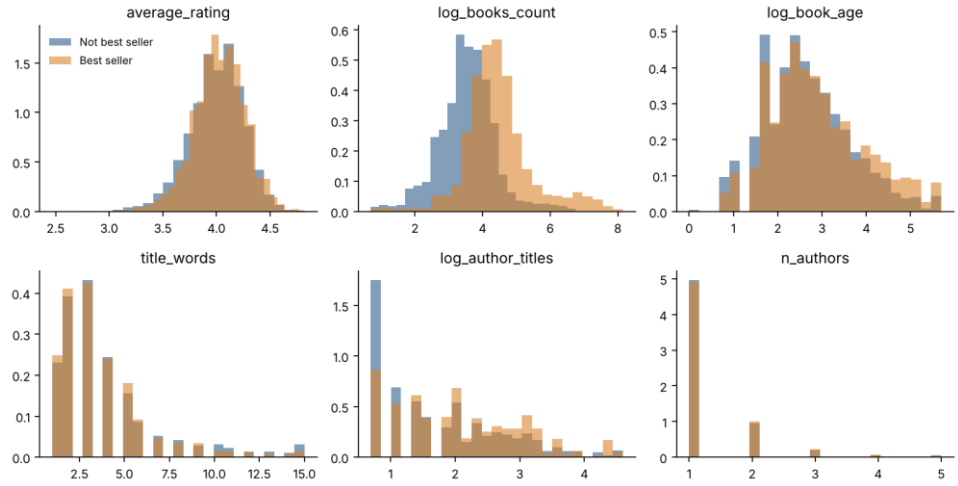


Figure 7: Distributions of the numeric features by class

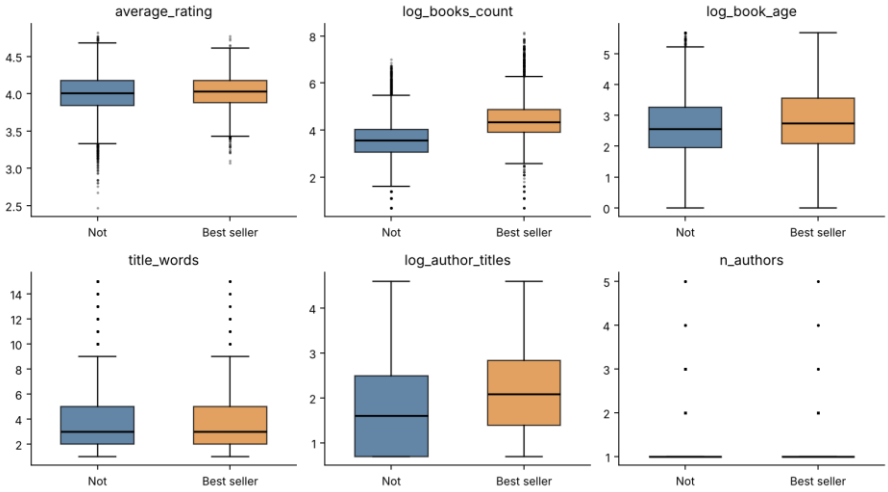


Figure 8: Box plots of the numeric features by class

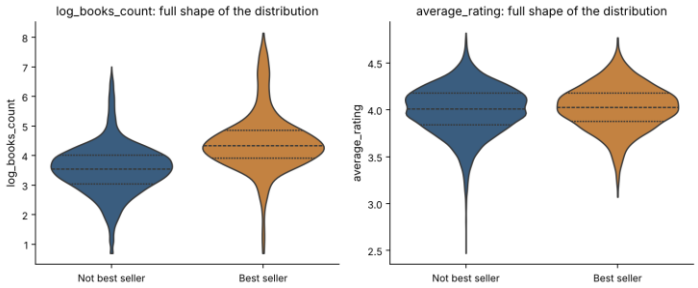


Figure 9: Violin plots of log editions and average rating by class

6.3 Yes/no features and the effect of time

First books in a series are best sellers more often than other books. Books where the known language is not English are at 5.9% compared to 20.3%, books with a subtitle are at 12.9% compared to 21.1%. The proportion of books that are best sellers declines over the years, from 31.9% to 16.7%, as older books have had more time to receive ratings, and as there are many more recent books in the data set (3,142 from the 2000s, 3,067 from the 2010s).

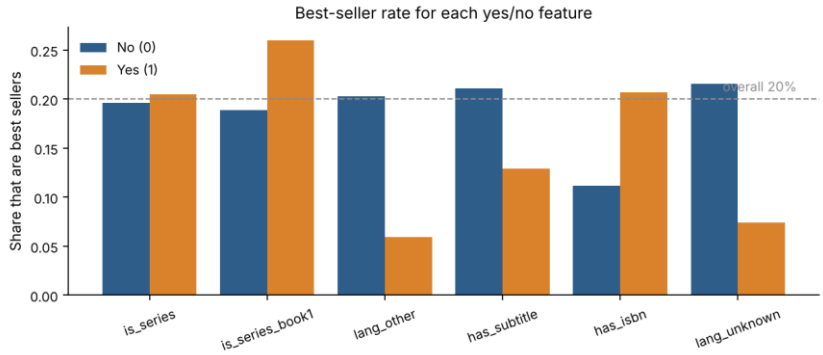


Figure 10: Best-seller rate for each yes/no feature

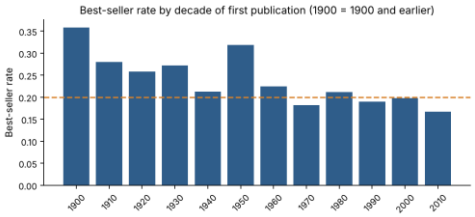


Figure 11: Best-seller rate by decade

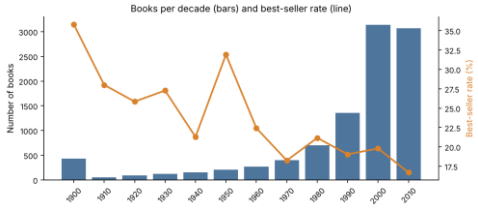


Figure 12: Books per decade and best-seller rate

The two effects interact. Within every era, first books in a series do better than standalone books, and later series books do worse than either (Figure 13). For books published before 1980 the rate is 35% for first books and 22% for later books; for the 2010s it is 22% and 15%.

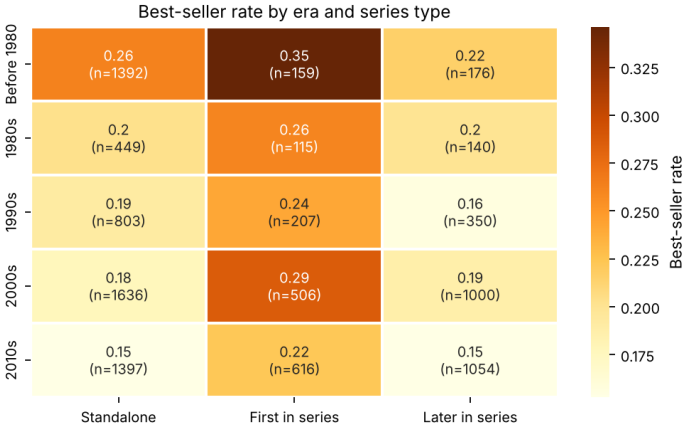


Figure 13: Best-seller rate by era and series type, with the number of books in each cell

6.4 Relationships between features

Editing and rating are more than one or the other. In the top five percent of the books rated with the highest number of editions, 64% of them are best sellers, while in the bottom five percent with the lowest number of editions, only 2% of the books are best sellers. The best-seller rate counts from 5.3% in the bottom 10% to 56.0% in the top 10% of editions. The hexbin chart depicts the same pattern in two dimensions: middle age books with many editions are the darkest, indicating the highest best-seller rates.

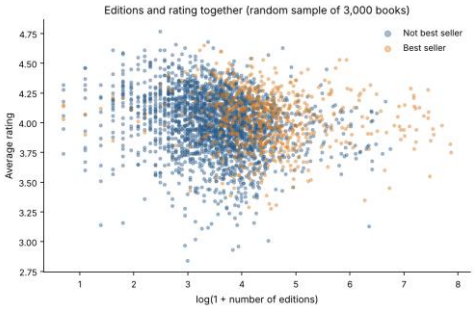


Figure 14: Editions against average rating

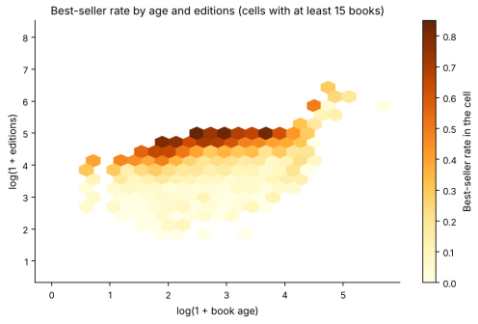


Figure 15: Best-seller rate by age and editions



Figure 16: Best-seller rate by decile of editions

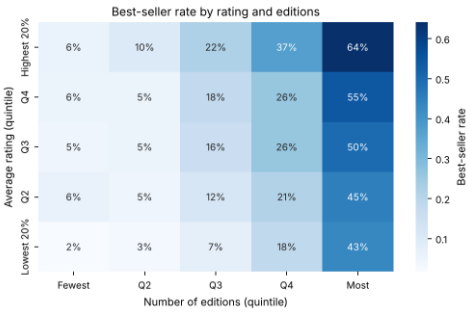


Figure 17: Best-seller rate by rating and editions

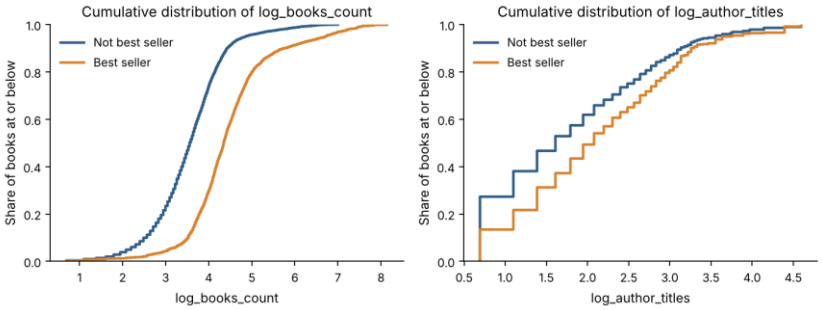


Figure 18: Cumulative distributions of editions and author catalogue size, by class

6.5 Correlation

Pearson correlation is used to measure straight-line association, and Spearman correlation is used to measure rank association. Both are in close agreement here. Author catalogue size (Pearson 0.38, Spearman 0.40) and unknown language (-0.11, -0.11) are the second and third strongest correlations, respectively. The correlation between average rating and the average of the ratings is low (0.04, 0.04). There is no strong correlation of any coefficient with another coefficient, which is good for interpreting coefficients individually.

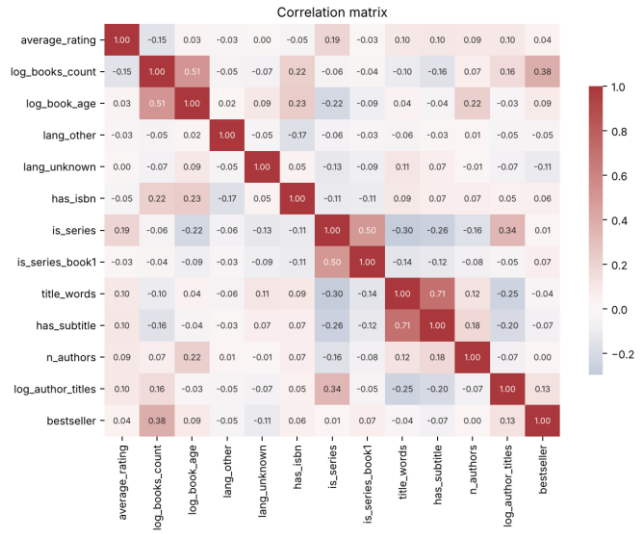


Figure 19: Correlation matrix of the features and the target

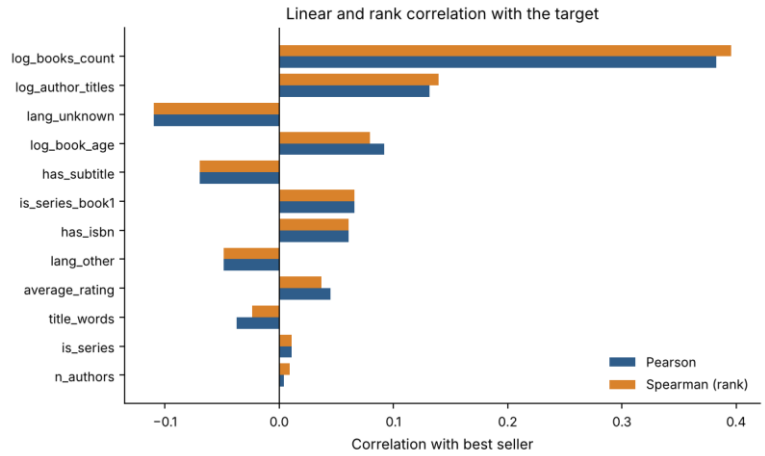


Figure 20: Pearson and Spearman correlation of each feature with the target

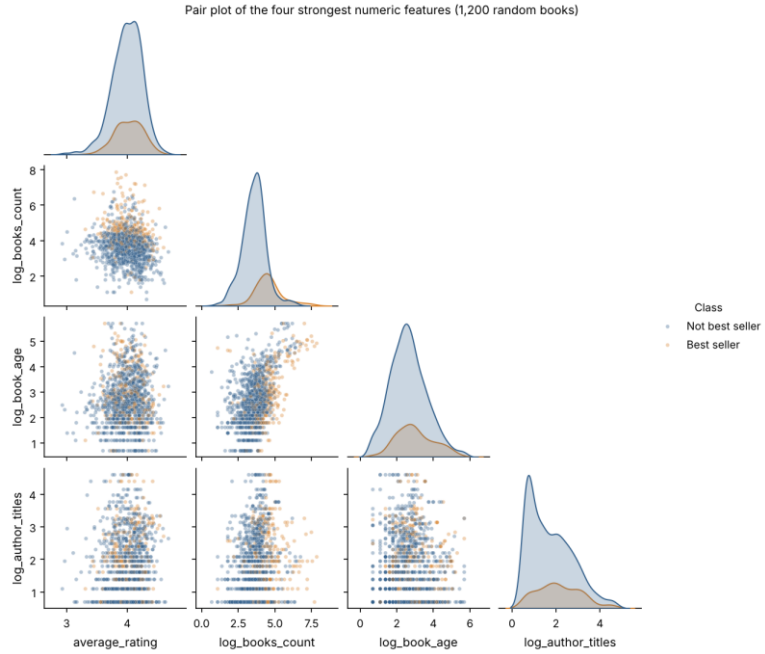


Figure 21: Pair plot of the four strongest numeric features (1,200 books)

6.6 Authors, star ratings and concentration

The importance of authors is not the same everywhere. The number of best-sellers is among the highest in the group of prolific authors, with Stephen King (80 books), the writing god, topping the list at 65% to J.D. Robb at 2.4%. The number of ratings for the two classes is almost equal: 39% of books have 5 stars, whereas 37% have other ratings. Rather there is a high concentration of attention: the best 1% of books account for 22.8% of all ratings, the best 10% account for 56.4%, and the best 20% (highest-traveled) account for 68.9% of all ratings.

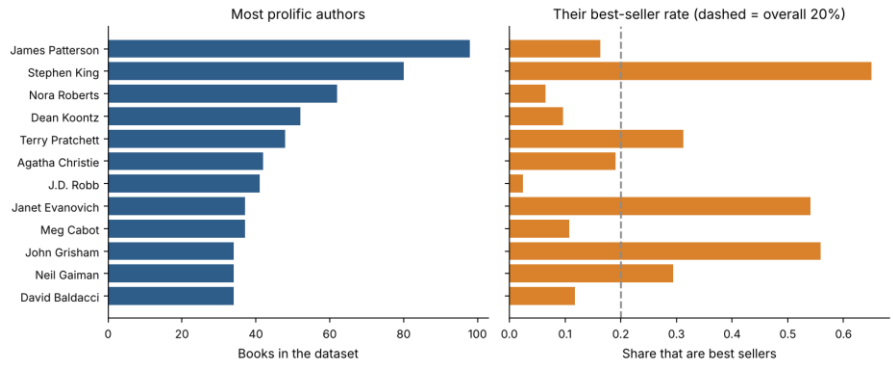


Figure 22: The most prolific authors and their best-seller rates

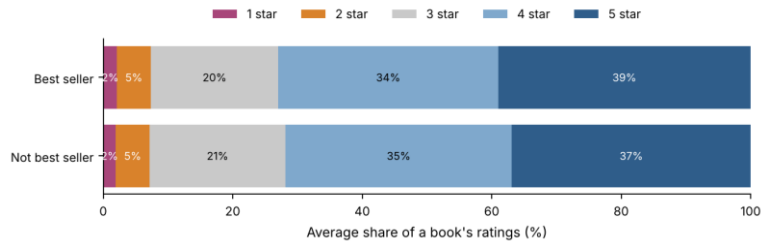


Figure 23: Share of star ratings, best sellers against other books (descriptive only)

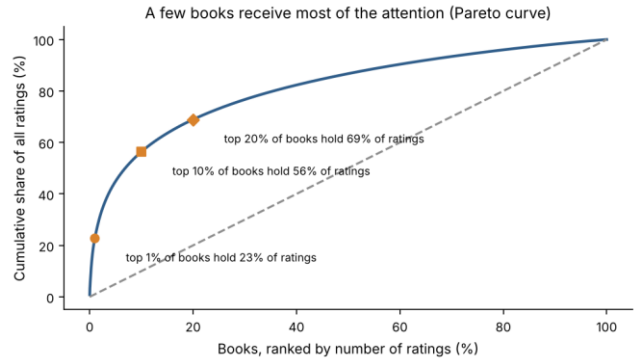


Figure 24: Concentration of ratings (Pareto curve)

6.7 Statistical tests

For each feature, the yes/no features were tested against the target using a chi-square test and numeric features were tested against the target using a Mann-Whitney U test. The effect size shown is the phi coefficient for chi-square and the rank-biserial correlation for Mann-Whitney, with the positive being the best sellers. The difference in numbers of books (10,000 almost any difference is statistically significant) is more important than the p-values.

Feature	Test	Statistic	Effect size	p-value	Reading
is_series	chi-square	1.1	0.011	0.292	No evidence of a difference
is_series_book1	chi-square	42.7	0.065	< 0.001	Very small effect
lang_other	chi-square	22.6	0.048	< 0.001	Very small effect
lang_unknown	chi-square	120.1	0.110	< 0.001	Small to moderate effect
has_isbn	chi-square	36.3	0.060	< 0.001	Very small effect

Feature	Test	Statistic	Effect size	p-value	Reading
has_subtitle	chi-square	47.6	0.069	< 0.001	Very small effect
average_rating	Mann-Whitney U	8,428,709	0.054	< 0.001	Very small effect
log_books_count	Mann-Whitney U	12,569,767.5	0.571	< 0.001	Large effect
log_book_age	Mann-Whitney U	8,913,960.5	0.114	< 0.001	Small to moderate effect
log_author_titles	Mann-Whitney U	9,595,471.5	0.199	< 0.001	Small to moderate effect
title_words	Mann-Whitney U	7,731,067.5	-0.034	0.018	Very small effect
n_authors	Mann-Whitney U	8,072,928.5	0.009	0.371	No evidence of a difference

Table 10: Univariate tests of each feature against the target

Editions stand out with a large effect (rank-biserial correlation 0.57). Is_series and n_authors show no evidence of a difference on their own, and title_words only a very weak one. They can still matter in combination with other features, which the multivariable model examines in section 7.

6.8 Checking the assumptions of logistic regression

Assumption	How it was checked	Result
Binary outcome, independent rows	One row per work; no duplicate identifiers	Met. Books by the same author may share traits, which is noted as a limitation.
Log-odds linear in each continuous feature	Observed log-odds in ten equal-sized bins of each feature	Met. Correlation between bin means and log-odds: editions 0.921, age 0.814, rating 0.776, author titles 0.797.
No severe multicollinearity	Variance inflation factor for every feature	Met. The largest VIF is 2.13 (limit of concern 5).
Enough events per variable	Events divided by variables in the training data	Met overall (145.45 per variable), but only 8 best sellers have a known non-English language, so that term is weakly supported.

Assumption	How it was checked	Result
No extreme influential cases	Leverage and Cook's distance	Met. See section 8.

Table 11: Assumption checks

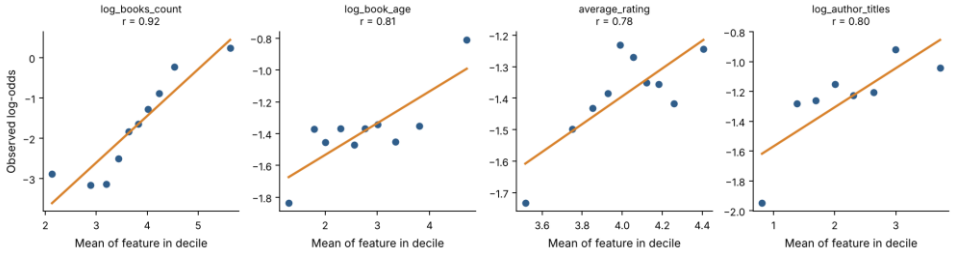


Figure 25: Observed log-odds in deciles of each continuous feature, with a straight-line fit

The straight-line pattern holds for all four continuous features, so a linear logit is reasonable. Age and rating show mild curvature, which a more flexible model could capture, but the loss from ignoring it is small.

Finding in the EDA	Decision it led to
80/20 class imbalance	Stratified split; F1, recall, PR-AUC and MCC beside accuracy; tuned decision threshold
Editions are extremely skewed (skewness 8.4)	log(1 + editions); likewise for age
Missing language carries information	Separate “unknown” language group
Series position and language matter	Three series and language flags with English as the reference group
Rating is weakly related on its own	Keep it, because it becomes stronger once editions and age are in the model (section 7)
Ratings define the target	Exclude every ratings-based column from the features
Low correlation among features	Coefficients can be read one at a time

Table 12: From findings to decisions

7 Feature selection

One of the 12 features (has_isbn) was removed using a backward elimination with $p < 0.01$. The 12-, 11- and 9-feature sets are equally predictive in cross-validation, so the choice of features is not much more predictive but more simpler.

How it was done (training data only)

1. Fit an un-penalised logistic regression with Newton-Raphson, coded from scratch (section 2.3).
2. Invert Hessian, take inverse Hessian as standard errors and then compute z = coefficient / standard error and a two-sided p-value (a Wald test).
3. Backward elimination: begin with all features and remove one with the highest p-value as long as p-value is above the cut-off.
4. Forward selection: begin with no variables and continually add the variable that has the smallest p-value if it falls below the threshold.).

Feature	Coefficient (per 1 SD)	p-value (full model)	VIF	Decision
log_books_count	1.690	< 0.001	1.55	Keep
log_book_age	-0.701	< 0.001	1.60	Keep
average_rating	0.457	< 0.001	1.17	Keep
is_series_book1	0.405	< 0.001	1.48	Keep
is_series	-0.327	< 0.001	1.92	Keep
lang_unknown	-0.287	< 0.001	1.05	Keep
lang_other	-0.244	< 0.001	1.05	Keep
log_author_titles	0.219	< 0.001	1.31	Keep
title_words	0.155	0.0020	2.13	Keep
n_authors	-0.132	< 0.001	1.11	Keep
has_subtitle	-0.141	0.0082	2.11	Keep
has_isbn	0.089	0.0280	1.13	**Drop** (p > 0.01)

Table 13: The full model: coefficients, p-values and VIF

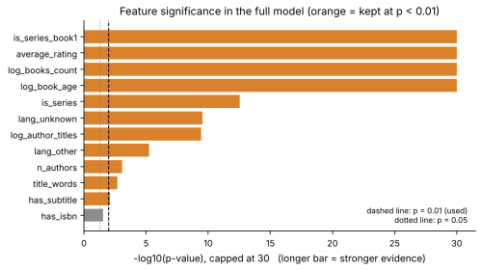


Figure 26: Significance of each feature in the full model

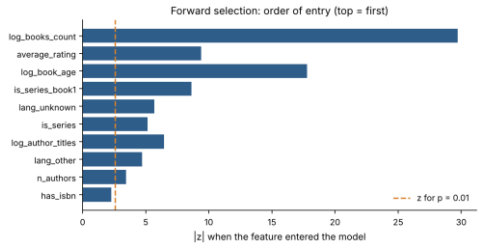


Figure 27: Forward selection: order of entry

Why p < 0.01 and not 0.05. If the significance level is p<0.05 there is no elimination at all as a tiny effect appears to be significant with 8,000 rows. The 12 tests are also performed simultaneously and so increase the risk of false positives (about 0.004 is the Bonferroni corrected value; Dunn, 1961). The 0.01 value eliminates the least significant feature and retains the others.

Backward and forward selection disagree, so cross-validation decides. Forward selection (p < 0.01) adds nine features, and never adds features such as title_words or has_subtitle which only become significant once the other features are added. There is also a suppression effect from forward selection: age enters in third, with |z| of 17.8; rating enters in third, with |z| of 9.4, but they reinforce each other and are pulling in opposite directions. The three sets do not significantly differ when tested by cross-validation::

Feature set	Features	CV ROC-AUC	CV PR-AUC
All features	12	0.8218	0.5997
Backward, 0.01 (11 features)	11	0.8219	0.5996
Forward, 0.01 (9 features)	9	0.8221	0.5983

Table 14: Feature sets compared by 5-fold cross-validation

Backward elimination is used because it considers the importance of each feature, taking into account all other features, and does not offer any loss of accuracy. Here an earlier coding of language (English flag and unknown flag) resulted in a VIF of 6.4 due to a very small reference group. All the VIFs after recoding were 2.2 or lower with the reference group being English.

One by one, adding features in forward order will yield less and less benefit. The cross-validated ROC-AUC of editions is 0.786; the rating is added, which boosts the performance to 0.810; the 8th feature increases it to 0.822, and the last 4 features don't improve anything further. The PR-AUC increases from 0.486 to 0.600 which is particularly significant in the first three steps.

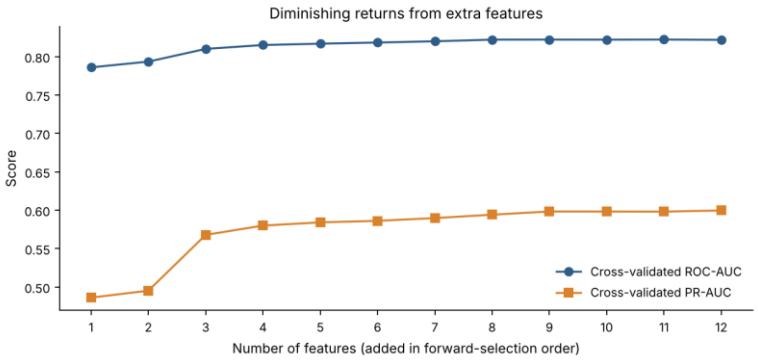


Figure 28: Cross-validated scores as features are added in forward-selection order

8 Model fit and diagnostics

To see how the 11-feature model fits the training data and to identify books that cause the model to fit badly, the model was fitted to the training data without penalty before the models were trained. Only training data is used in these checks.

8.1 Overall fit

Statistic	Value	Reading
Likelihood-ratio test against the intercept-only model	chi-square 1877.93 on 11 df, p < 0.001	The features together predict far better than the base rate alone
McFadden pseudo R-squared	0.235	Moderate; pseudo R-squared is not comparable with linear R-squared (McFadden, 1974)
Nagelkerke pseudo R-squared	0.331	About a third of the variation explained (Nagelkerke, 1991)
AIC and BIC	6152.51 and 6236.36	For comparing models fitted to the same data
Hosmer-Lemeshow test (10 bands)	chi-square 92.40 on 8 df, p < 0.001	Rejects a perfect fit; see below
Events per variable	145.45	Far above the guide of 10 (Peduzzi et al., 1996)

Table 15: Fit statistics of the 11-feature model (training data)

The Hosmer-Lemeshow test (Hosmer and Lemeshow, 1980) says that a perfect fit is rejected and it is perfectly honest to tell the truth. The test is very sensitive (8,000 rows) and the table indicates where the misfit is found. There are 32 books in the lowest band; the model predicted 8.1 books in that band, so the actual count is within about 20 books of the model prediction in both directions. The other nine bands are within about 20 books of the expected number in each band, in either

direction. The model is well-calibrated for most books in the set, is a poor predictor of best sellers in the set, and a good predictor of almost certain failures of books in the set with few editions. This is also true on the test set in section 10.

Band of predicted probability	Books	Observed best sellers	Expected by the model
1 (lowest)	800	32	8.1
2	800	26	25.5
3	800	39	45.1
4	800	48	66.3
5	800	90	92.2
6	800	106	123.5
7	800	149	163.9
8	800	203	222.5
9	800	339	315.9
10 (highest)	800	568	537

Table 16: Hosmer-Lemeshow bands: observed against expected best sellers

8.2 Influential books

Cook's distance (Cook, 1977) indicates the change in fit (i.e. the RMSSE) if one book is removed. The largest value is 0.0117, much smaller than the value of 1 at which one case is typically a problem. By the stricter rule of 4/n (0.0005), 453 of 8,000 books (5.7%) exceed the line, which is typical for a sample this large. The five most influential books are Dreamcatcher, Alias Grace, Cruel & Unusual, Cujo, and Christine. The maximum leverage is 0.041, and 701 books have leverage more than twice the average.

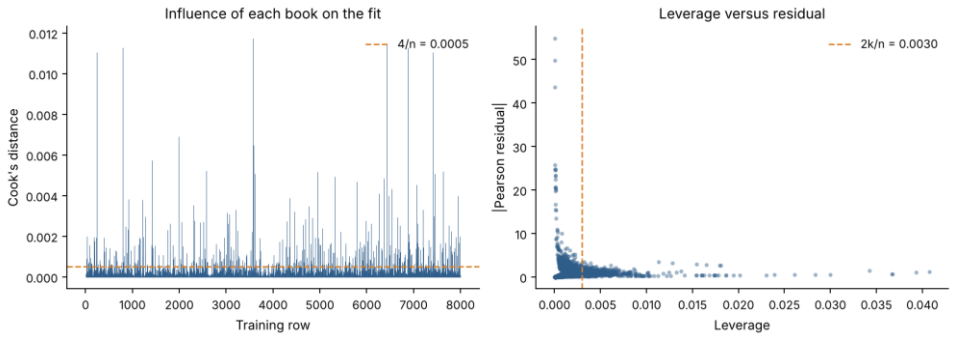


Figure 29: Cook's distance and leverage for each training book

Robustness check and an honest caveat. The removal of the 80 most influential books (this is the top 1% of books, with 72 of them being best sellers) and refitting changes no coefficient sign. All other coefficients change by less than 0.41, except the intercept. The difference is that the non-English term (lang_other) has no best sellers, as all of the 152 known non-English books in the training data are among the list of influential books, so after removal there are no best sellers and the coefficient tends to be negative (complete separation). The finding of lower scores for books that are not written in English is based on a small number of events (8) and should be regarded with suspicion.

9 Training the model from scratch

The model uses only plain NumPy: a sigmoid function, a loss function, and gradient descent that adjusts the coefficients until the loss does not decrease anymore; scikit-learn is used just for splitting the data, calculating metrics and double-checking the result. The steps below are matched with a block of the commented code in the notebook.

Step 1: Prepare the data. The data is cleaned and then split 80/20, with stratification, so that there are 8,000 training books and 2,000 test books, and each book has exactly 20% best sellers. Standardisation of features is done based on training statistics only (mean=0, standard deviation=1). This ensures that all features are put on one scale, causing quick convergence of gradient descent and prevents information from the test set from infiltrating.

Step 2: Turn features into a probability. Make steps turn into a likelihood. The model takes all the features ($z = w \cdot x + b$) adds them up and passes them through the sigmoid function, which is used to compress the value into the range of 0 to 1. The result is the probability that the book is a best seller.

$$p = 1 / (1 + e^{-(w \cdot x + b)})$$

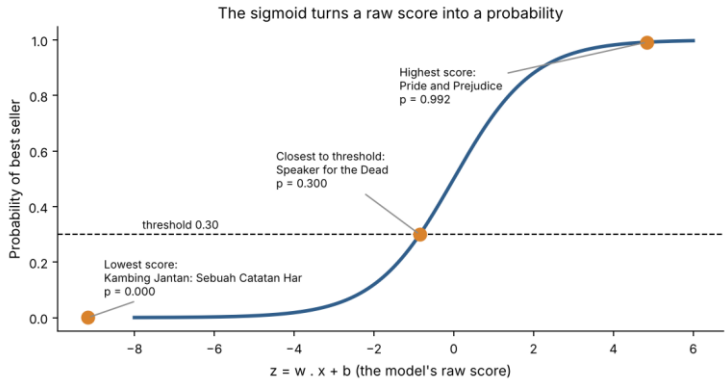


Figure 30: The sigmoid turns a raw score into a probability, with three real test books

This is illustrated in three real test books. Pride and Prejudice has a raw score of 4.84, and a probability score of 0.992 (and it is a best seller). Speaker for the Dead is just on the edge, a score of -0.85 with a probability of 0.300, and another best seller. The probability of the lowest scoring book is 0.0001.

Step 3: Measure how wrong the model is. The loss function is called the binary cross-entropy. It has a high penalty for confident mistakes and there is only one best answer to gradient descent and it is convex. Large weights are reduced by an optional L2 penalty (λ) and errors on the rare best-seller class are penalized more due to the optional class weight.

$$L = -(1/n)\ \Sigma\ [\ y \cdot \log(p) + (1 - y) \cdot \log(1 - p)\ ] + (\lambda/2) \cdot ||w||^2$$

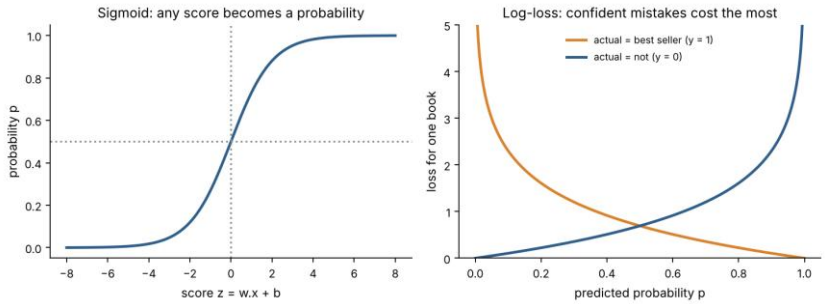


Figure 31: The sigmoid function and the log-loss for one book

Step 4: Work out which way to move. The gradient indicates the change in loss with each change in weight. The simple form for logistic regression is the average of (predicted probability - actual outcome) multiplied by each feature.

$$dL/dw = (1/n)\cdot X'(p - y) + \lambda\cdot w \qquad dL/db = \text{mean}(p - y)$$

Step 5: Take a step and repeat. The forward pass, the calculation of the loss, the calculation of the gradient, and the updating of the weights is performed in each iteration. Training is stopped when the number of iterations reaches the maximum or if the loss is decreased by less than 0.000000001. The sigmoid is written to prevent overflow, a very small number ensures the logarithm is not undefined, and the intercept is not penalised for safety..

A 2 feature illustration helps to make an idea visible. All weights start at zero, and in 90 steps, gradient descent moves from a loss of 0.693 (as bad as guessing the base rate of zero) to a loss of 0.419 and weights of 1.10 on log editions and 0.30 on average rating, which are close to the weights that converge to at loss of 0.439 (1.13 and 0.32). The right decision surface is the same and the black line is the 0.30 threshold.

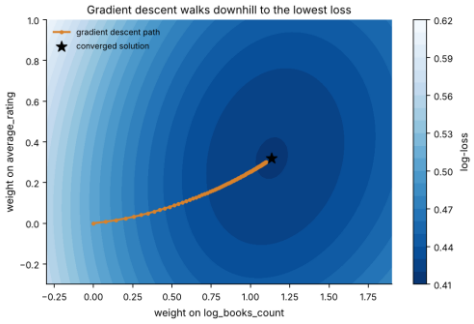


Figure 32: Gradient descent on the loss surface

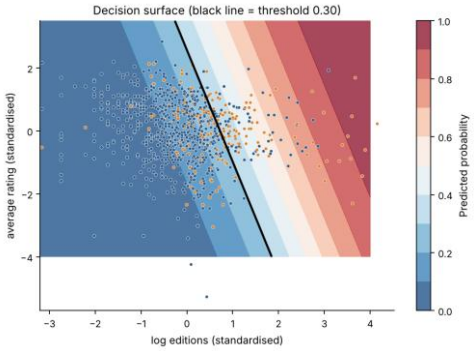


Figure 33: Decision surface for two features

Step 6: Set the hyper-parameters. The defaults for model 1 are: learning rate 0.1, 1000 iterations, no L2 penalty, no class weights, all 12 features and a decision threshold of 0.5. Section 11 tunes these.

The learning rate will determine the rate of descent. After 100 iterations the loss is 0.600 with a rate of 0.01, 0.423 with 0.1 and 0.385 with 0.5. After 1000 iterations, the three fastest rates have converged to a loss of 0.383, and 0.01 still has a loss of 0.422. The optimum has been obtained to machine precision (shown by the length of the gradient in the right-hand panel, which falls to 9.5e-8 at 0.5 rate). Here also a rate of 1.0 is converging, but steps larger than this can overshoot on other data..

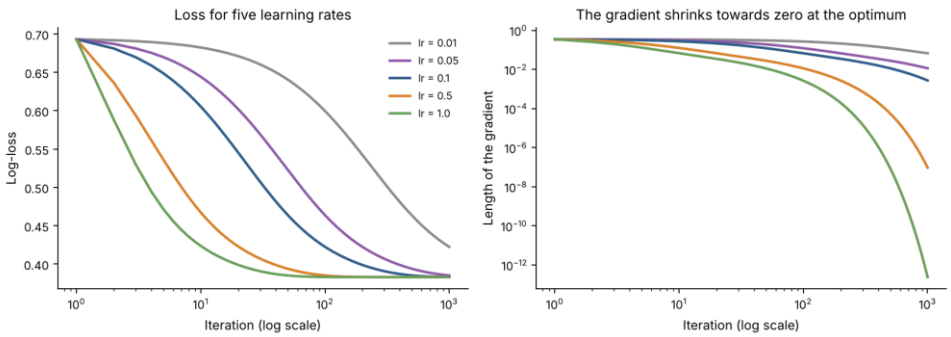


Figure 34: Loss and gradient length for five learning rates

Step 7: Check that training worked. The loss curve is smooth and flat, indicating that gradient descent converged. The Model 1 takes all 1,000 iterations whilst the Model 2 (which has a higher learning rate) terminates at 518. The model is not over-fitting as it achieves a similar score for training data and testing data (ROC-AUC 0.823 and 0.813, accuracy 0.841 and 0.844.

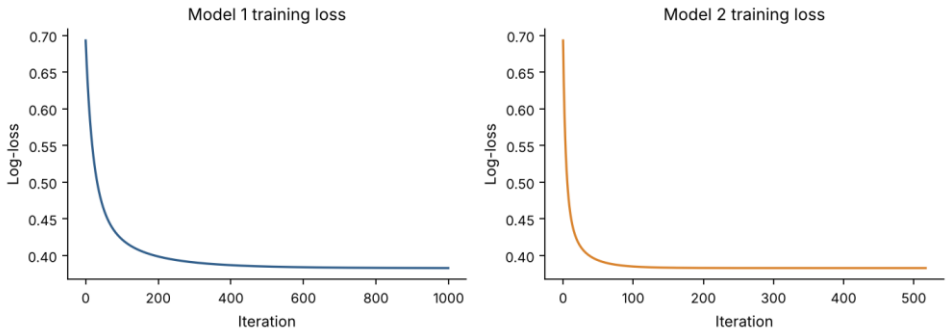


Figure 35: Training loss of Model 1 and Model 2

Step 8: Prove the maths is right. The same data was used to compare the scratch model with the unpenalised LogisticRegression from scikit-learn. The coefficient difference in the largest is 0.0024, the coefficient difference of the Newton-Raphson is 0.0024, and the difference of the test ROC-AUC is 0.8125 for both.

Regularisation. The larger the L2 penalty, the closer all the coefficients will approach zero. The coefficient of the editions goes from 1.69 to 0.046 and the penalty is quite large. A light penalty will only change the coefficients slightly, hence there is no advantage in tuning a light penalty as in section 11.

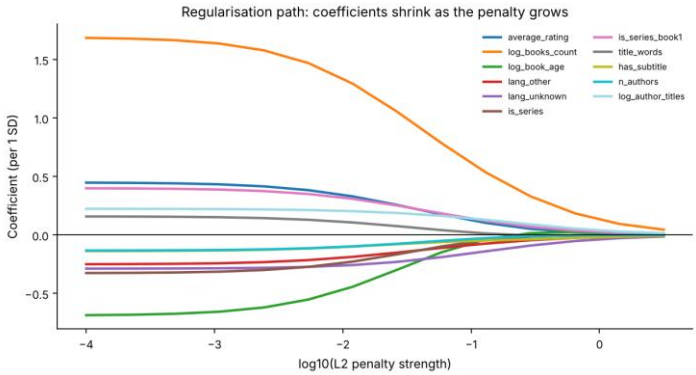


Figure 36: Coefficients as the L2 penalty increases

10 Evaluation and interpretation

On 2,000 unseen books the final model has a ROC-AUC of 0.813 (95% bootstrap interval 0.786 to 0.838), finds 57.5% of best sellers and is right on 53.5% of the books it flags. The data in this section is from a test set locked for use, and only employed once.

10.1 Metrics chosen and why

Metric	Question it answers	Why it matters here	Model 1	Model 2
Accuracy	Share of all predictions that are right	Easy to read, but 0.80 is the trivial score at 80/20	0.844	0.815
Precision	Of books flagged, how many are best sellers?	The cost of backing a book that flops	0.754	0.535

Metric	Question it answers	Why it matters here	Model 1	Model 2
Recall	Of real best sellers, how many are found?	The cost of missing a hit	0.323	0.575
F1	Balance of precision and recall	One score for the rare class	0.452	0.554
ROC-AUC	Chance a random best seller is ranked above a random non-best-seller	Ranking quality at every threshold; 0.5 is guessing	0.813	0.813
PR-AUC	Ranking quality focused on the rare class	Guessing scores 0.20	0.611	0.610
MCC	Correlation of prediction and truth, using all four confusion-matrix cells	Robust to imbalance	0.424	0.438
Log-loss	Quality of the predicted probabilities	The training objective	0.392	0.393

Table 17: Evaluation metrics on the test set

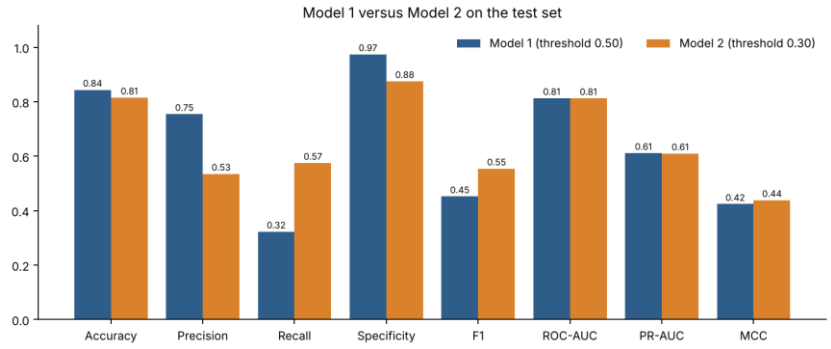


Figure 37: Model 1 against Model 2 across the main metrics

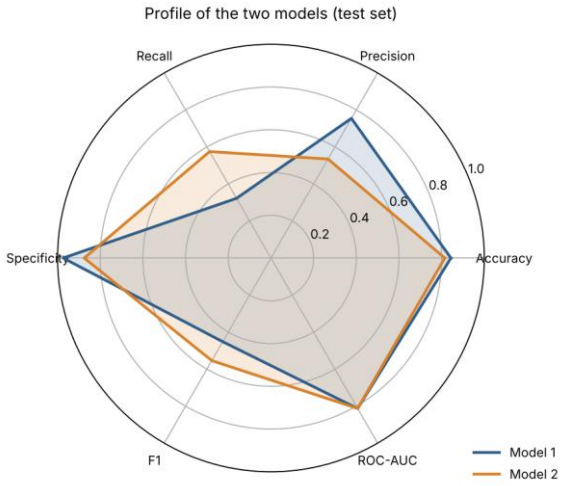


Figure 38: Profile of the two models on six metrics

Reading the table: ranking quality is good and identical for both models. About 81% of the time a random best seller will be outranked by a random non-best seller, while guessing will have a PR-AUC of only 0.20, three times smaller than the PR-AUC of 0.61. The difference lies in the models' different thresholds for deciding whether to flag a book: Model 1 is conservative (high precision, low recall) and Model 2 is less conservative (low precision, high recall). The radar chart demonstrates

that Model 2 has a tradeoff where precision, specificity and accuracy are sacrificed for the benefit of recall and F1.

10.2 Confusion matrices and what the predictions look like

Model Only 42 false alarms with Model 1 finding 129 of the 400 best sellers. Model 2 missed 170, but got 230, and the cost was 200 false alarms. There is no over-fitting from the view of the model performance score on the training and test data: Model 2 ROC-AUC score is 0.823 on the training data and 0.813 on the test data.

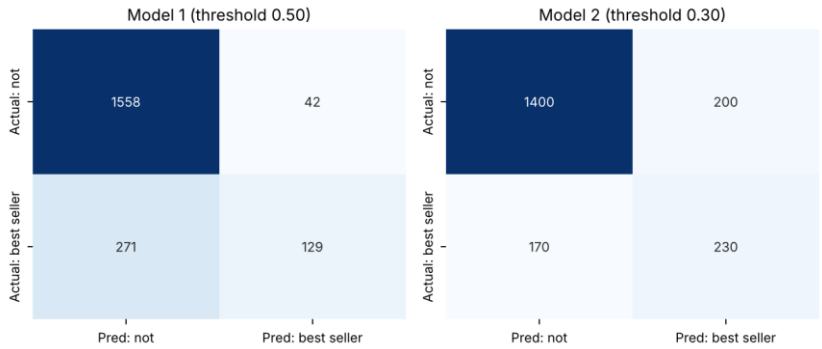


Figure 39: Confusion matrices of Model 1 and Model 2

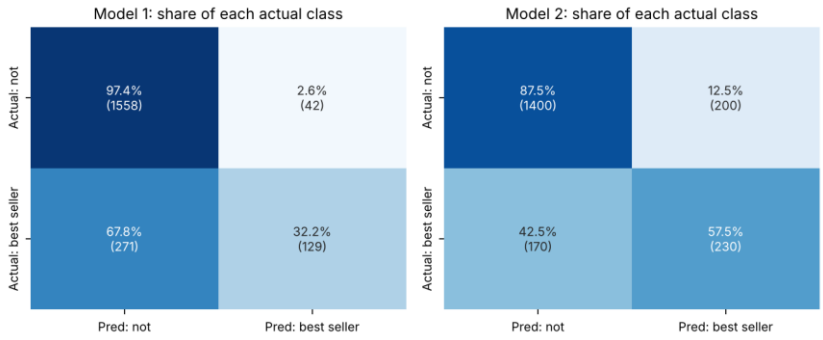


Figure 40: Confusion matrices as a share of each actual class

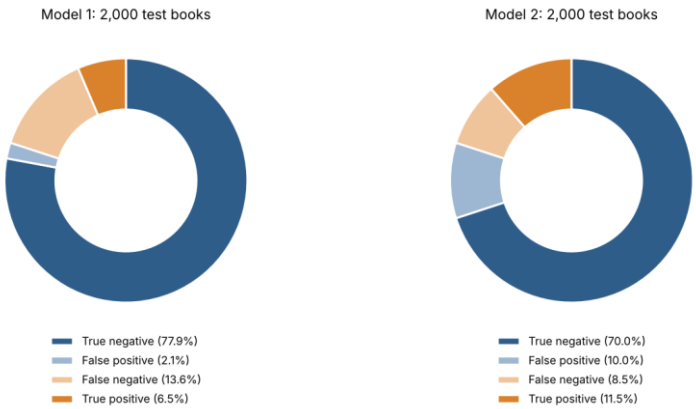


Figure 41: Outcome of every test prediction, as a share of the 2,000 books

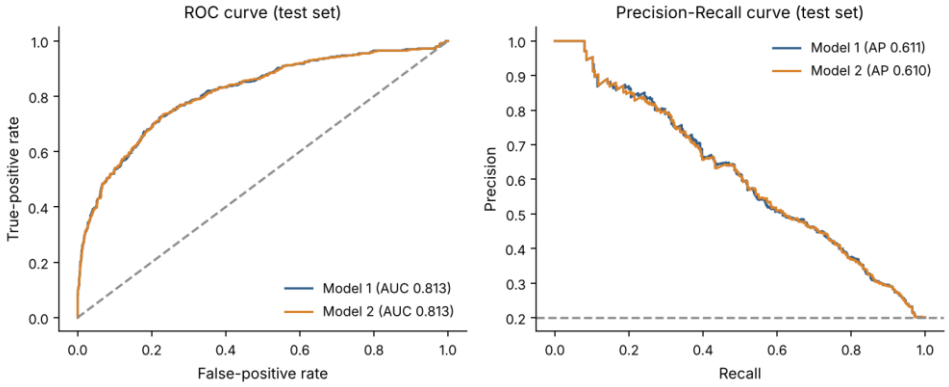


Figure 42: ROC and precision-recall curves on the test set

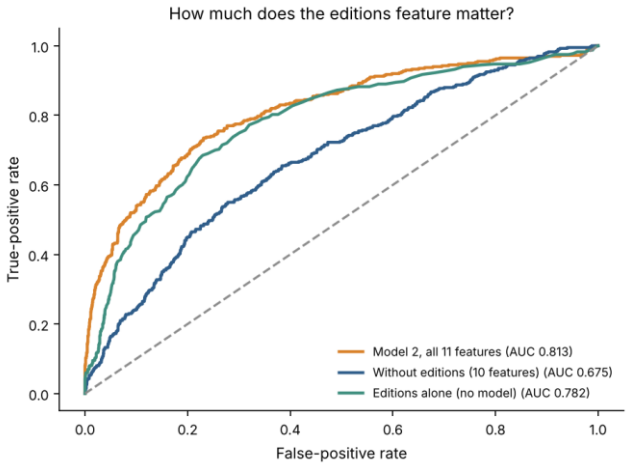


Figure 43: How much the editions feature matters

The editions carry the majority of the signal. If there is no model, but only editions, the AUC is 0.782. The full version goes to 0.813. The AUC is 0.675 (PR-AUC 0.346) without editions. The other ten add some real, but limited value on top of editions.

10.3 How well separated and calibrated are the scores?

The median predicted probability is 0.37 for best sellers, and 0.10 for other books. The two distributions have a Kolmogorov-Smirnov statistic of 0.504 with a score of 0.21. The calibration is good: the calibration curve is close to the diagonal, and the Brier score is 0.117 compared to 0.160 for a model that always predicts the base rate (Brier, 1950; Niculescu-Mizil and Caruana, 2005). The same is found in the decile chart: 66% and 73% are best sellers in the top decile. The only area where there is a visible difference is in the lowest decile, which the model predicts 1% while 7% are best sellers, similar to the Hosmer-Lemeshow result in section 8.

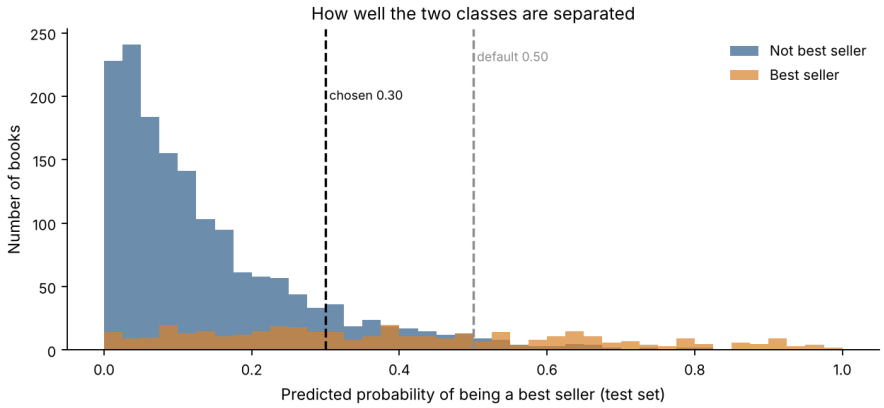


Figure 44: Predicted probabilities for best sellers and other books

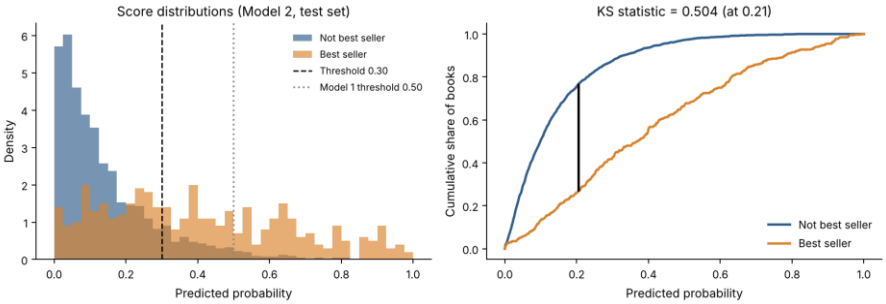


Figure 45: Score distributions and the KS statistic

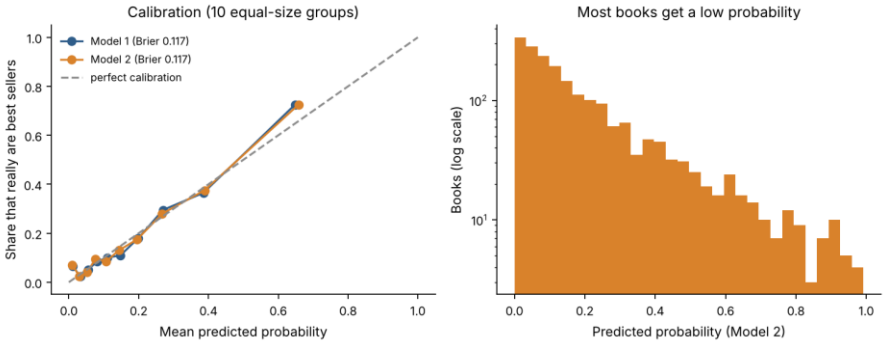


Figure 46: Calibration curve and how often each score is used

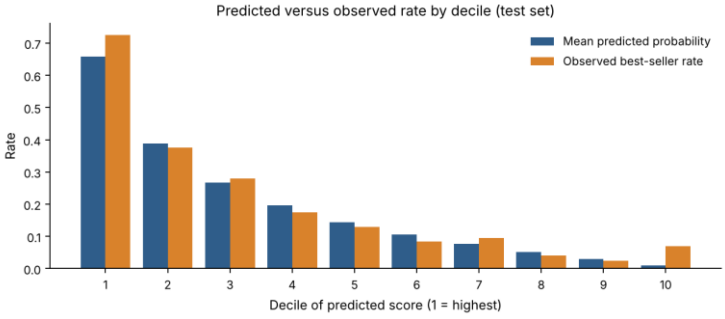


Figure 47: Predicted against observed best-seller rate by decile of score

10.4 Value of the ranking to a publisher

A publisher can't read all books so the ranking is important. Reviewing only the top 10% of books by score finds 36.2% of all bestsellers (a lift of 3.62 over random picking); the top 20% finds 55% (lift 2.75); the top 30% finds 69% (lift 2.3); and the top half finds 84.2% (lift 1.69).

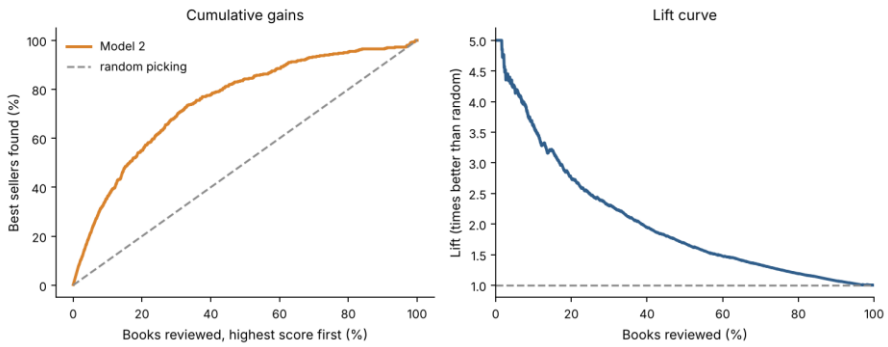


Figure 48: Cumulative gains and lift charts

10.5 Stability: would the result hold up?

The averaged ROC-AUC of the model over five cross validation folds is 0.822 with a standard deviation of 0.0075 (range from 0.807 to 0.829), which is the same for the four set-ups compared. The bootstrap interval for the test AUC, 0.786 to 0.838, is about ±0.026 wide. The learning curve demonstrates that the AUC of the validation data increases from 0.816 with 640 training data to 0.822 with 6,400 training data, while the AUC of the training data decreases from 0.840 to 0.823. The curves cross indicating that it is not overfitting and more of the same type of data will not be of much help, a better feature would.

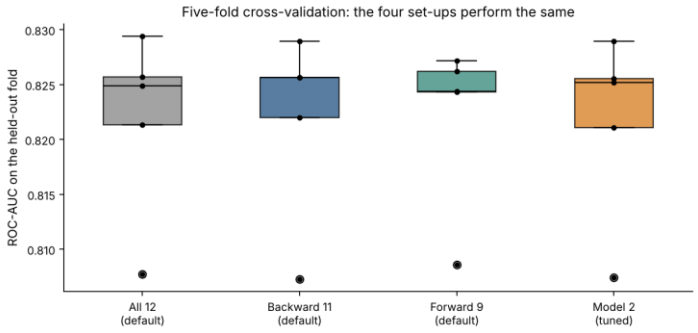


Figure 49: Cross-validated ROC-AUC for four set-ups

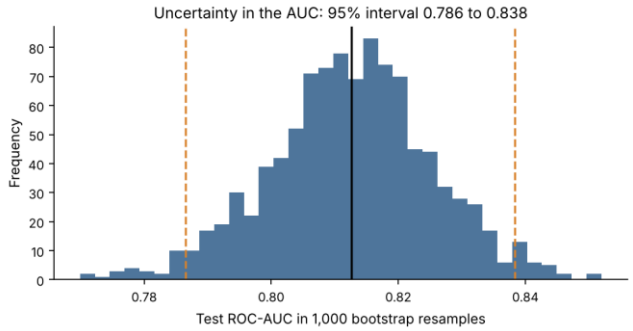


Figure 50: Uncertainty in the test ROC-AUC (1,000 bootstrap resamples)

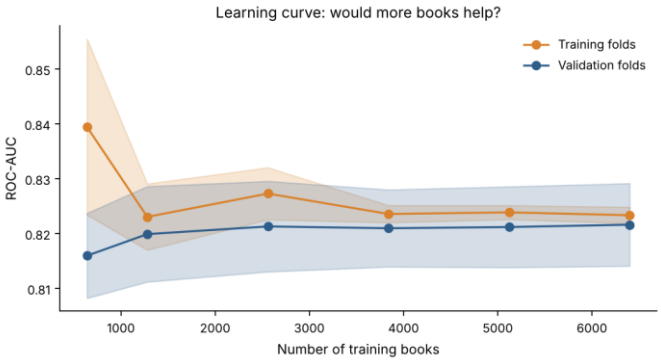


Figure 51: Learning curve: training and validation ROC-AUC against training size

10.6 Does it work equally well for every kind of book?

Group	Books in test set	Best sellers	ROC-AUC	Recall	Precision
Language: English	1,748	380	0.815	0.600	0.548
Language: Other known	34	3	n/a	0.000	n/a
Language: Unknown	218	17	0.663	0.118	0.143
Series: First in series	335	79	0.844	0.797	0.553
Series: Later in series	552	104	0.770	0.375	0.494
Series: Standalone	1,113	217	0.825	0.590	0.540
Era: Before 1980	336	99	0.781	0.697	0.556
Era: 1980s	122	21	0.765	0.476	0.556
Era: 1990s	264	57	0.816	0.579	0.611
Era: 2000s	657	122	0.807	0.500	0.550
Era: 2010s	621	101	0.850	0.564	0.463

Table 18: Performance by subgroup of the test set (AUC shown only where there are at least 10 best sellers)

Performance is not uniform. Books in English-language series (AUC 0.815, recall 60%) and first books in a series (AUC 0.844, recall 80%) are best suited to the model. It is significantly weaker on books of an unknown language (AUC 0.663, recall 12%) and fails to recall most of the best sellers of the later years of the series (recall 38%). It didn't detect any of the 3 best sellers that use a language other than English, but the number of these sellers is too low to be able to judge. By era, the AUC ranges from 0.765 for the 1980s to 0.850 for the 2010s. One threshold cannot apply to all of these groups.

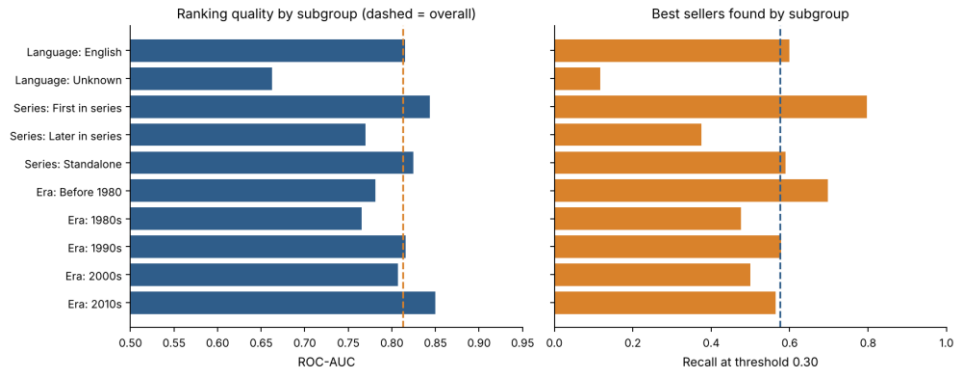


Figure 52: ROC-AUC and recall for each subgroup of the test set

10.7 What the model relies on

Feature	Odds ratio	95% interval	Plain meaning
log_books_count	5.43	4.87 to 6.05	Strongest signal: roughly every 2.6-fold rise in editions multiplies the odds by about 5.4
average_rating	1.57	1.46 to 1.69	A quarter-star higher rating raises the odds by about 57%
is_series_book1	1.49	1.39 to 1.61	First books in a series do better than later ones
log_author_titles	1.25	1.17 to 1.34	Prolific authors have more best sellers
title_words	1.17	1.06 to 1.29	Slightly longer titles help a little
n_authors	0.88	0.81 to 0.95	More credited people lowers the odds slightly
has_subtitle	0.87	0.78 to 0.97	A subtitle lowers the odds slightly
lang_other	0.78	0.70 to 0.86	Known non-English books do less well than English ones (based on few events; see section 8)
lang_unknown	0.75	0.69 to 0.82	Unknown language does less well than English
is_series	0.72	0.66 to 0.79	Later books in a series do less well than standalone books (odds of about half, 0.52, for the yes/no flag itself)
log_book_age	0.50	0.46 to 0.55	Older books have lower odds once editions are held fixed

Table 19: Odds ratios per one-standard-deviation increase, other features held fixed

An odds ratio above 1 raises the odds of being a best seller and below 1 lowers them. The raw data is more suggestive of the older books being best sellers (correlation +0.09), which is due to the fact that books that are classics have many editions. The model partitions these two effects and is responsible for the negative odds ratio for the effect of age. If a book is part of a series, the two series flags combined provide a bit more than 1.5 x the odds of a standalone book.

There is a second view that does not involve the coefficients, which is provided by permutation importance. One feature at a time shuffling, fall in test AUC, editions: 0.294, about ten times as many as any other feature; rating: 0.034, about ten times as many as any other feature; the two series flags: 0.015 and 0.015; and author catalogue: 0.010. All the other features cost less than 0.01.

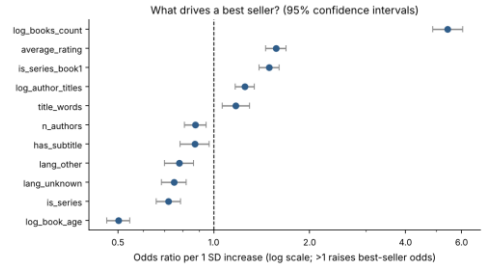


Figure 53: Odds ratios with 95% confidence intervals

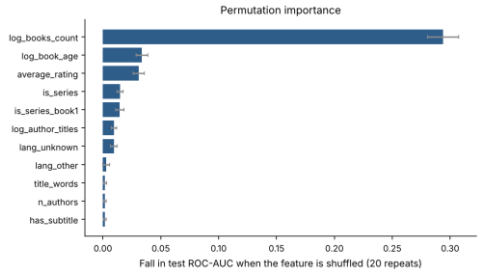


Figure 54: Permutation importance on the test set

Three individual cases show how the features combine. Each bar is a feature's push on the log-odds for one book.

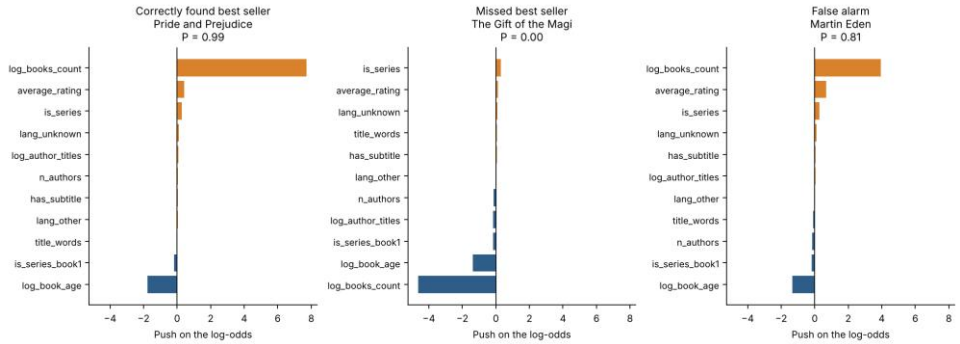


Figure 55: Why the model scored three books as it did

10.8 Where the model goes wrong

The Public domain classics that have been published many times have the highest chances with ratings lower than the cut-off but in the range of 77% to 81% (Beyond Good and Evil, Martin Eden, King Solomon's Mines, The Case-Book of Sherlock Holmes, Kidnapped, and others), with 260 to 849 editions in the top eight. The most missed best sellers (The Gift of the Magi, The Rise and Fall of the Third Reich, Midnight Sun, Skeleton Crew, The Gathering Storm) had just one to 13 editions, so the model had less than a 1% chance for each. The model works well for reading “many editions” as “popular”, which is average for these two groups, and is wrong for them. When compared to equally sized groups of subjects by editions, the model matches reality reasonably closely, but under-performs in the lowest and highest groups.

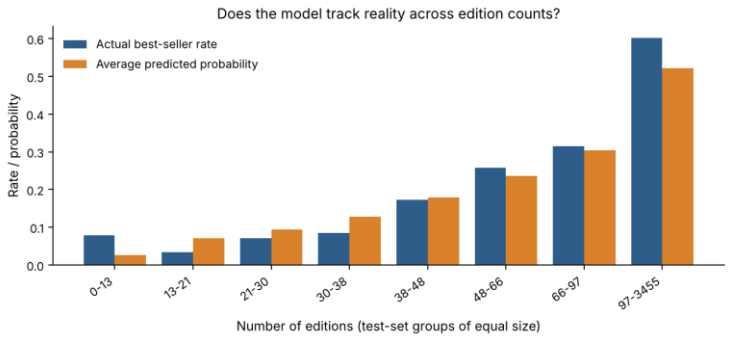


Figure 56: Actual best-seller rate against average predicted probability, by group of editions

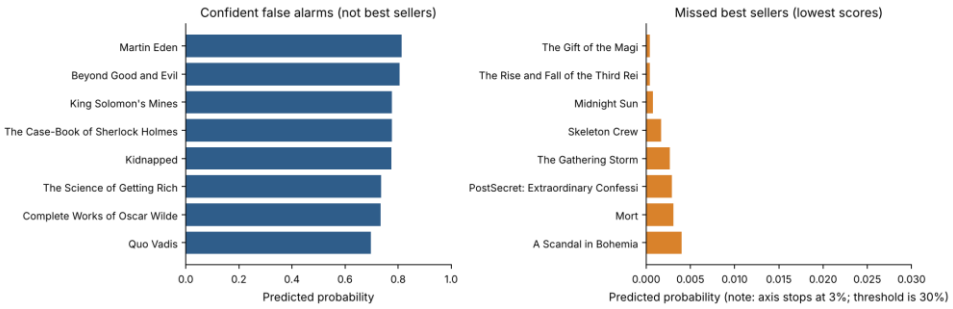


Figure 57: The most confident false alarms and the most-missed best sellers

11 Retraining: what changed and what it achieved

Retraining increased the recall to 57.5% and F1 to 0.554, while maintaining the ranking power of 0.4.With no loss in ranking power, retraining increased recall to 57.5% and F1 to 0.554. The majority of the gain was from lowering the decision threshold from 0.50 to 0.30, the rest of the tuning had little effect. One of the primary problems with model 1 was that it failed to capture 68% of best sellers, stemming from the fact that an 80/20 problem would have a 68:22 split, with the 68% being considered a “fussy” problem. Every change listed below addresses a weakness and is supported by cross validation of the training set only.

Shortcoming of Model 1	Change in Model 2	Evidence
Recall of only 0.323	Decision threshold tuned from out-of-fold predictions: highest F1 at 0.30	Cross-validated F1 0.568; test recall 0.575
A weak feature kept (has_isbn, p = 0.028)	Backward elimination at p < 0.01 leaves 11 features	CV ROC-AUC 0.8219 against 0.8218 with all 12
Untuned learning rate and penalty	Grid of 4 learning rates × 4 L2 strengths × 2 class-weight settings (32 models), 5-fold CV	Best: learning rate 0.5, no L2 penalty, CV PR-AUC 0.6007. The default (0.1) scores 0.6006, so these settings barely matter
Class imbalance	“Balanced” class weights tested against none	At each option's own best threshold, CV F1 is 0.565 with weights and 0.568 without, so no weights are used

Table 20: Shortcomings of Model 1 and the changes made

The final model trains with learning rate 0.5 and stops after 518 iterations because the loss has stopped improving. Because no class weights are used, its probabilities are not distorted by re-weighting (test log-loss 0.393, the same as Model 1), so they can be compared across books.

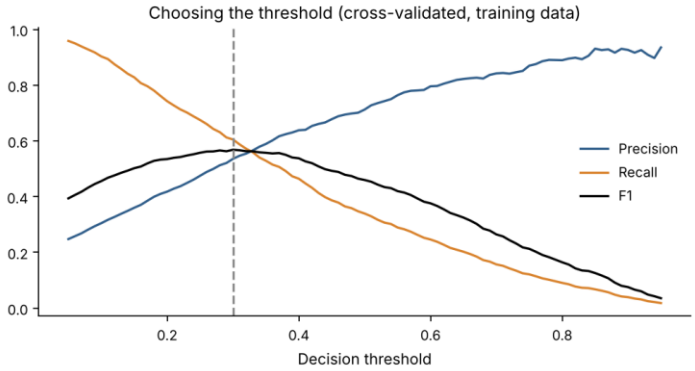


Figure 58: Choosing the decision threshold from cross-validated predictions

11.1 The threshold is a business decision

The value of 0.30 is one of many possible F1-optimal thresholds. If the threshold is increased, it will reduce false alarms, but it will pass fewer best sellers; if the threshold is decreased, then it will pass more best sellers but will have more false alarms (Figure 60). When the two mistakes are the same cost, the best threshold is 0.46 and when the miss is twice as much, the best threshold is 0.37 and three times is 0.28 and five times is 0.18. These optima are based on predictions made from out of fold training data, and are not from the test data.

Threshold	Best sellers found	False alarms	Best sellers missed
0.2	296	392	104
0.3	230	200	170
0.4	177	102	223
0.5	133	45	267
0.6	100	21	300

Table 21: Test-set outcomes at five thresholds (Model 2)

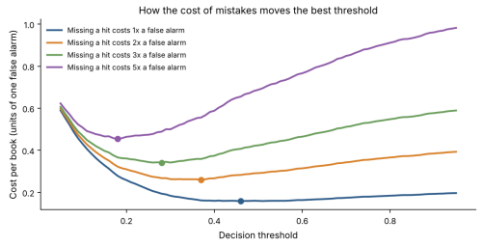


Figure 59: Cost of mistakes against threshold

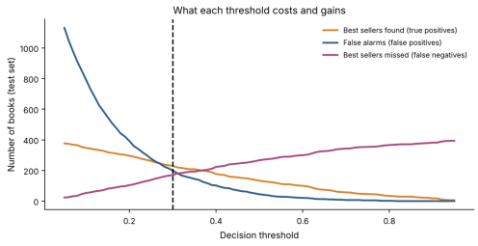


Figure 60: Found, missed and false alarms by threshold

11.2 Hyper-parameter tuning

The cross-validated PR-AUC is just between 0.539 and 0.601, and the loss surface isn't very noisy so the learning rate and a small L2 penalty are barely distinguishable. A large penalty (0.1) is a little harsh, as it reduces the useful coefficients a bit too much. Balanced class weights reduce the PR-AUC a bit (from 0.60 to 0.59), since they alter the loss without adding information. The complete grid is in Appendix B.

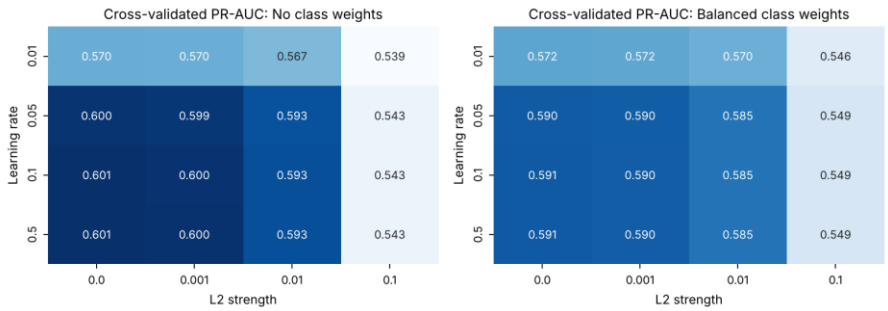


Figure 61: Cross-validated PR-AUC for every learning rate and L2 strength, without and with class weights

Option	Learning rate	L2	CV PR-AUC	Best threshold	CV F1 at that threshold
No class weights	0.5	0	0.6007	0.30	0.5682
Balanced weights	0.5	0	0.5906	0.61	0.5649

Table 22: Class-weight options compared by cross-validation

12 Discussion

Answering the research question. Yes, yes, with limits. The model's rank is well above chance (ROC-AUC 0.81, KS 0.50) and the probabilities of the model are generally well calibrated for most books. It's able to identify 57% of best sellers at the selected threshold and approximately one book flagged out of every two is a true best seller. It's a useful improvement over the random approach of picking a book from the top if the publisher is looking for the first book to review: the top 5% of the books receives 55% of the hits.

What drives success in this data. The number of editions outweighs. Editions can be, in part, a by-product of success, as popular books are reprinted, so the model can be, in part, learning that popular books are popular. The less actionable findings are the smaller effects with the editions kept fixed: that is, the odds are higher for books that are first in a series, by established authors with higher average ratings, and lower for books with subtitles or later in a series..

Methodological choices that mattered. The three decisions that made the most difference were:The three decisions that made the biggest difference were: A first, treating accuracy as the goal would have favoured those with a 2-3 record, that is, Model 1, which makes two of the top three best sellers. Choosing instead the F1 and the recall revealed the problem (Saito and Rehmsmeier, 2015; Chicco and Jurman, 2020). Second, the improvement was not due to the learning algorithm, but due to the decision threshold, which is consistent with the idea that the moving threshold is a simple and effective way to solve the problem of imbalance (He and Garcia, 2009). Third, selecting a more stringent p value cut-off and cross validation prevented over-reading of small differences between feature sets.

Ethical and fairness considerations. The model is a positive reinforcement for what is already popular. The size of the author catalogue and number of editions are features that grant advantages to the established authors and popular titles, and would leave a house of publishing new authors, translated titles, or niche titles out of consideration if they depended on this model. There is also lower performance for books that are unknown language, and for later books in a series (section 10.6). The data does not include any personal information about readers, but the ratings are based on a subset of book buyers who self selected to participate in Goodreads. The model should be used alongside rather than instead of human editors, and should be regularly checked for bias against under-represented groups of authors.

Comparison with what theory predicts. These results are consistent with those of the literature for imbalanced data. However, while accuracy seemed to hide an inferior recall, precision-recall measures gave the real story, and calibration plots demonstrated that the class-agnostic trained model was able to maintain useful probabilities (Niculescu-Mizil and Caruana, 2005). This is a common result for a small number of variables (k) and a large number of observations (n) as in this example.

13 Limitations and recommendations

Use the model as a screening tool for books that already have a track record. It is not a pre-launch sales forecast.

Limitations

- •Proxy label. The word best seller does not mean best selling; it only refers to the number of Goodreads ratings and not all Goodreads users are book buyers.
- Only popular books. Each book in the data is already popular, so the model does not differentiate between hits and flops, but between very popular and moderately popular books.
- Editions are somewhat based on success. If you don't include the editions feature, the ROC-AUC decreases to 0.675. Most false alarms are caused by public-domain classics with hundreds of editions.
- Missing drivers. The data is not available for genre, price, marketing spend, cover, pre-orders and the author's past sales – this limits the accuracy of the data to approximately 0.81 AUC.
- Sparse evidence for non-English books. The coefficient is fragile, because only 8 of 152 known non-English books in the training data are best sellers.
- Fitting poorly at the bottom. The Hosmer-Lemeshow test rejects the perfect fit because the model under-predicts books it rates as near-certain failures that are best sellers.
- Association, not cause. The more editions or a first in series, the more likely a book is to be a best seller, but there are no rules that dictate the number of editions that will make a book a best seller.
- One random split, repeated authors. Since books were not divided by time, we have not tested the aging of the model. On either side of the split you'll find books by the same author, which may do a little of a flattery service to the author-catalogue feature.

Recommendations

1. Use Model 2 to shortlist titles for reprints, promotion or rights decisions, and review each flag by hand. About one flagged book in two is a true best seller.
2. Set the threshold by cost. Use 0.30 for balanced coverage; decrease it if the production would lose points if it didn't hit; increase it if budget is limited for promotion (section 11.1).
3. Add genre, price, marketing spend and author sales history before using the model on unpublished titles.
4. 4Collect more data on non-English and independent titles before trusting the model for them.
5. Re-train annually and monitor recall and precision for new books.

Future work. Natural next steps would be a split based on time, to see if the slightly curvy nature of age and rating is significant (curved effects, for example splines or gradient-boosted trees, could be tested to see if the mild curvature in the age and rating is important); author-level grouping in the validation split; and a label generated based on actual sales..

14 Conclusion

A logistic regression reimplemented and trained with gradient descent and validated on scikit-learn is able to predict best sellers with an ROC-AUC of 0.81 on unseen books. The first model was accurate, but failed to correctly classify the majority of the best sellers; the threshold was shifted from 0.50 to 0.30, increasing recall from 32% to 57% and F1 from 0.45 to 0.55. The number of editions, the rating, and the series position and author catalogue are the strongest drivers. The model is reasonably well calibrated over most of its range, most useful for English-language and first-in-series books, and should be used as a signal for human review, not to predict sales. The notebook that accompanies this report can be run to reproduce every number in this report.

NB : APPENDIX AFTER REFERENCE LIST

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Appendix A Glossary

Term	Meaning
Accuracy	Share of all predictions that are correct
AUC (ROC-AUC)	Area under the ROC curve; the chance a random positive is scored above a random negative
Backward elimination	Start with all features and remove the least useful one at a time
Best seller (this study)	A book above the 80th percentile of ratings count
Calibration	Whether predicted probabilities match observed frequencies
Class weights	Extra importance given to errors on the rare class during training
Cook's distance	A measure of how much one case changes the fitted model
Cross-validation	Repeatedly training on part of the data and testing on the rest to estimate performance
F1 score	The balance (harmonic mean) of precision and recall
Forward selection	Start with no features and add the most useful one at a time
Gradient descent	An iterative method that moves coefficients downhill on the loss surface
Learning rate	The size of each gradient-descent step
Leverage	How unusual a book's feature values are compared with the rest
Log-loss (cross-entropy)	The loss that logistic regression minimises
MCC	Matthews correlation coefficient; a balanced single score from the confusion matrix
Odds ratio	The factor by which the odds change when a feature rises by one unit
Precision	Of the books flagged, the share that are best sellers
Recall (sensitivity)	Of the real best sellers, the share that are found
Regularisation (L2)	A penalty that shrinks coefficients to reduce over-fitting
Threshold	The probability above which a book is flagged as a best seller
VIF	Variance inflation factor; a measure of collinearity between features

Table 23: Glossary of terms

Appendix B Full hyper-parameter grid

All 32 settings tried, in order of cross-validated PR-AUC (5-fold, training data only). The first row is the setting chosen for Model 2 before the class-weight comparison.

Learning rate	Iterations	L2 penalty	Class weights	CV PR-AUC	CV ROC-AUC
0.5	2000	0.0	none	0.6007	0.8216
0.1	2000	0.0	none	0.6006	0.8217
0.5	2000	0.001	none	0.5999	0.8218
0.1	2000	0.001	none	0.5999	0.8218

Learning rate	Iterations	L2 penalty	Class weights	CV PR-AUC	CV ROC-AUC
0.05	2000	0.0	none	0.5996	0.8219
0.05	2000	0.001	none	0.5989	0.8219
0.5	2000	0.01	none	0.5933	0.8220
0.1	2000	0.01	none	0.5932	0.8219
0.05	2000	0.01	none	0.5927	0.8219
0.5	2000	0.0	balanced	0.5906	0.8213
0.1	2000	0.0	balanced	0.5906	0.8213
0.05	2000	0.0	balanced	0.5904	0.8213
0.5	2000	0.001	balanced	0.5899	0.8213
0.1	2000	0.001	balanced	0.5899	0.8213
0.05	2000	0.001	balanced	0.5897	0.8213
0.5	2000	0.01	balanced	0.5846	0.8208
0.1	2000	0.01	balanced	0.5846	0.8208
0.05	2000	0.01	balanced	0.5846	0.8208
0.01	2000	0.0	balanced	0.5723	0.8187
0.01	2000	0.001	balanced	0.5720	0.8186
0.01	2000	0.0	none	0.5702	0.8185
0.01	2000	0.001	none	0.5698	0.8184
0.01	2000	0.01	balanced	0.5695	0.8181
0.01	2000	0.01	none	0.5667	0.8177
0.5	2000	0.1	balanced	0.5492	0.8118
0.1	2000	0.1	balanced	0.5491	0.8118
0.05	2000	0.1	balanced	0.5491	0.8118
0.01	2000	0.1	balanced	0.5464	0.8109
0.5	2000	0.1	none	0.5427	0.8095
0.1	2000	0.1	none	0.5427	0.8095
0.05	2000	0.1	none	0.5426	0.8095
0.01	2000	0.1	none	0.5389	0.8084

Table 24: The full hyper-parameter grid

Appendix C Supporting tables

Decile of score (1 = highest)	Books	Best sellers	Mean predicted	Observed rate
1	200	145	0.658	0.725
2	200	75	0.389	0.375
3	200	56	0.267	0.280

Decile of score (1 = highest)	Books	Best sellers	Mean predicted	Observed rate
4	200	35	0.196	0.175
5	200	26	0.143	0.130
6	200	17	0.107	0.085
7	200	19	0.077	0.095
8	200	8	0.052	0.040
9	200	5	0.030	0.025
10	200	14	0.010	0.070

Table 25: Predicted against observed best-seller rate by decile (Model 2, test set)

Cost of a miss relative to a false alarm	Best threshold	Cost per book (units of one false alarm)
1 times	0.46	0.159
2 times	0.37	0.260
3 times	0.28	0.341
5 times	0.18	0.454

Table 26: Best threshold for different costs of mistakes (out-of-fold training predictions)

Appendix D Code map and reproducibility

Notebook section	What it does	Report section
0 to 1	Imports, settings, workflow diagram, dataset audit	3
2	Cleaning and feature engineering	5
3, 3.9, 3b	Exploratory analysis, composition, spread, relationships, assumption checks, tests	6
4	Stratified split and scaling	9 (step 1)
5	Logistic regression from scratch: sigmoid, gradient descent, Newton-Raphson, Wald p-values	2, 9
6, 6b	Backward and forward selection; fit statistics and influence diagnostics	7, 8
7	Model 1 (baseline)	9, 10
8	Cross-validated tuning, class weights and threshold; Model 2	11
9, 9.9, 9b	Evaluation, calibration, gains, stability, subgroups, importance, error analysis	10
10	Validation against scikit-learn	9 (step 8)

Table 27: Map from notebook sections to report sections